

Acoustic shaping of custom lens surfaces

Geometric constraints, dual traction, distributed sources and stability
with Cartesian ovoids and simultaneous front–back shaping

Acoustic Freeform Lab
Private research manuscript

25 September 2026 — Version 4.0

Abstract

We organize acoustic lens shaping as a sequence from prescribed geometry to required traction, continuous acoustic synthesis, finite source realization, and maintained operation. The multiplier field enforcing the target surface is its required normal traction. Extending this variational inverse to both lens faces exposes their shared volume, pressure and acoustic coupling. A continuous quadratic integral operator maps distributed sources to loads; array placement, frequency and amplitudes restrict and optimize that map. A local analytic pressure-node construction illustrates conditional realizability and supplies a necessary carrier-displacement bound. Cartesian ovoids specialize the general geometry without defining the actuator architecture. The accuracy specification is a maximum absolute cycle-mean error of ten nanometres on each driven-liquid face; instantaneous motion is a separate diagnostic. Reachability, formation from the unforced equilibrium, and stability under three-dimensional disturbances remain distinct conditions. The results are conditional mathematical deductions and a verification framework; they do not establish ten-nanometre performance of a physical apparatus.

Contents

1	Problem, scope and conventions	5
1.1	Cylindrical geometry and data left as arguments	5
1.2	Three meanings of a stable optical result	6
1.3	Model hierarchy and mathematical status	6
1.4	Notation used throughout	7
2	Fluid mechanics and the load required by a target	7
2.1	A continuum reference problem	7
2.2	Wetting is part of the problem data	8
2.3	Variational derivation of the required static traction	9
2.4	What a finite pulse can leave behind	9
3	Geometric constraints, dual traction, and two lens faces	10
3.1	The geometric multiplier is a traction field	10
3.2	Front and back in one apparatus	11
3.3	The non-optical annulus is a design variable	12
4	Acoustic transfer, radiation traction and temporal memory	12
4.1	When a harmonic reduction is justified	12
4.2	The sign and content of radiation traction	12
4.3	Impulses are causal histories, not instantaneous intensities	13
5	Distributed sources and their forward operator	14
5.1	Source distributions precede an array	14
5.2	An explicit transmission problem	14
5.3	Input power is an independent consistency condition	15
5.4	Cylinder symmetry does not impose source symmetry	16
6	Local trace synthesis and its geometric cost	16
6.1	Two-fluid traces and a pressure node	16
6.2	Conditional local fields and volume forces	16
6.3	A necessary bound for strict geometric accuracy	17
7	Attainable loads and a regularized inverse	17
7.1	Continuous quadratic load operators	17
7.2	Necessary source-effort and sign bounds	18
7.3	Why arbitrary source functions do not imply arbitrary loads	18
7.4	A well-defined regularized fixed-target problem	19
7.5	A second-order entry that needs no neighboring target	19
7.6	What changes when the surface is also unknown	20
7.7	Finite arrays approximate a source space	21
8	Finite-array restrictions and coherent-load relaxation	21
8.1	A linear wave response produces a quadratic load	21
8.2	Feasibility, coherence and a useful relaxation	22

9	Continuous source synthesis and placement design	23
10	An explicit source adjoint with interface conditions	24
10.1	A traction objective on a fixed interface	24
10.2	The adjoint has jumps even when pressure does not	25
10.3	Spatial source gradients and necessary optimality	25
10.4	A limiting boundary check and extension to the full model	26
11	Stability of a maintained interface	26
11.1	A stationary force balance is only the starting point	26
11.2	Linearize the actual base flow in three dimensions	27
11.3	Capillary stiffness and acoustic feedback	27
11.4	Feedback must have the necessary authority and observations	28
11.5	What is needed to extend a linear certificate locally	29
12	A two-interface inertial formation diagnostic	29
13	Determine the load before synthesizing emitters	30
13.1	Two full-domain loads with an explicit sign convention	30
13.2	A conditional three-dimensional stability result for ideal loads	31
13.3	Formation requires a load history, not just its endpoint	31
13.4	Verification gates before returning to source design	32
14	Variational load inversion with coupled viscous fluids	32
14.1	The load is a multiplier of a physical evolution constraint	32
14.2	What changes when inertia and additional forcing are retained	33
14.3	A forward test that does not prescribe its answer	33
15	Formation as constrained optimal control	34
15.1	State, control and a dimensionally defined objective	34
15.2	Constraints carry physical meaning	35
15.3	Adjoint equations give a general derivative method	35
15.4	Robustness uses one realizable policy	36
16	Cartesian geometry as a target family	36
16.1	An unsquared optical-path construction	36
16.2	The four axial conjugate parameters	37
16.3	A common intermediate conjugate for the two faces	38
16.4	Geometric compatibility and limits of arbitrariness	38
16.5	Specialization of the general inverse problem	39
16.6	From four optical parameters to a conditional source map	39
17	The two-face ten-nanometre specification	40
18	Carrier-aware numerical inverse and acceptance gates	41
18.1	Coherent source steps with explicit carrier constraints	41
18.2	Sampled peak majorants	42
18.3	A relaxation diagnostic, not a substitute waveform	42

18.4	Why the maximum spot also requires slope control	42
18.5	Static acoustic feedback in the precision inverse	43
18.6	Required independent verification	44
19	Regimes, unresolved physics and the route to a design	44
19.1	Scales that determine which equations may be reduced	44
19.2	Physical effects that may set the attainable optical error	45
19.3	A theoretical specification before any numerical experiment	46
19.4	Numerical experiments will test stated predictions	46
19.5	What this formulation establishes and leaves open	47
A	Derivations and consistency checks	48
A.1	Optical path, branch selection and Cartesian curvature	48
A.2	Area, curvature and the pressure multiplier	48
A.3	Dissipation after actuation has ended	48
A.4	Quadratic sensitivities and local load authority	49
A.5	Discrete moving-graph acoustic tangent	49
A.6	Normalization and sign audit	51
B	Optional optical-performance diagnostics	51
B.1	Optical objectives and a slope-sensitive identity	51
B.2	Fast optical modulation	52
	References	53

1 Problem, scope and conventions

The design question is the following: given a smooth target interface inside a cylinder, which admissible acoustic source distributions can form it and maintain it with specified accuracy and stability? The target is initially an arbitrary surface. Its compatibility with the fluid volume and boundaries, the required stress, the range of realizable acoustic loads, and its dynamic stability are separate conditions. Cartesian optical surfaces enter only after these general conditions have been formulated.

Existing observations establish deformation by radiation pressure and the importance of the interaction between the wave and its evolving interface [1, 2]. Time-dependent deformations have also been measured under localized ultrasound [3]. These results motivate the present formulation, but do not establish arbitrary surface controllability or optical precision for an unspecified apparatus.

1.1 Cylindrical geometry and data left as arguments

Let t denote physical time, and let $T_f > 0$ denote a chosen formation horizon. The position vector is $\mathbf{x} = (x_1, x_2, z) \in \mathbb{R}^3$. Define a reference cylinder of radius $R_c > 0$ and height $H_c = z_t - z_b > 0$ by

$$\mathcal{D}_c = \{\mathbf{x} : x_1^2 + x_2^2 < R_c^2, z_b < z < z_t\}. \quad (1.1)$$

Here z_b and z_t are the bottom and top coordinates. The cylinder fixes the container geometry, not a symmetry of the state. An interface, source field or perturbation may depend on all three coordinates. A connected fluid region $\Omega^-(t)$ is surrounded, at least across its optical interface, by a second fluid region $\Omega^+(t)$. The superscript $-$ labels the lens material and $+$ the adjoining fluid; either may be the incident optical medium. A gas is included as a possible adjoining fluid, without replacing it by a vacuum or a pressure-release boundary at this stage.

Let $\Gamma(t)$ be the deformable fluid–fluid interface and $\Gamma_{\text{opt}}(t) \subseteq \Gamma(t)$ its clear optical part. Other boundaries may support the liquid, radiate acoustic energy, or couple to elastic solids. The actuator boundary, when actuation is imposed on a boundary, is denoted by Γ_a . The contact line, when one exists, is $\mathcal{C}(t) = \partial\Gamma(t)$ on a supporting solid. The two phases partition the fluid-filled part of the cylinder; additional immersed bodies require their geometry and boundary conditions as explicit data. Closed free interfaces have no contact line. A graph over a circular disk is one admissible representation, not a requirement; overhangs need a general surface description.

When acoustic motion is prescribed on a wall, $\partial\mathcal{D}_c$ is its reference or cycle-mean position. The full waveform problem uses the moving wall, and its linearization transfers the condition to that reference surface. An exactly fixed impermeable wall cannot also have a nonzero normal velocity. Any open boundary or volume source must declare its mass and energy exchange.

The unit normal $\mathbf{n}$ points from Ω^- to Ω^+ . Define the identity tensor $\mathbf{I}$, tangential projector $\mathbf{P}_\Gamma = \mathbf{I} - \mathbf{n} \otimes \mathbf{n}$, surface gradient $\nabla_\Gamma = \mathbf{P}_\Gamma \nabla$, and signed total curvature $\kappa = \nabla_\Gamma \cdot \mathbf{n}$. Thus a spherical lens-fluid region has positive curvature. The symbol $\otimes$ denotes a tensor product. For a quantity with two interface traces, define its jump as $\llbracket f \rrbracket = f^+ - f^-$. These choices fix the signs of all interfacial forces.

The supplied physical data must include the fluid constitutive laws, body force, external boundaries, wetting law or pinning constraint, conserved fluid masses, actuator response and admissible command set. A device-independent formulation leaves those data as arguments of the problem; it cannot eliminate their influence on the answer. In particular, optical conjugates alone cannot determine electrical voltages. No particular aperture, fill height, immersion fluid, source count or operating frequency is assigned. Arbitrary sources means arbitrary functions

within a declared physical admissible set, not independent prescription of pressure and velocity at every interface point.

1.2 Three meanings of a stable optical result

Let $\Gamma_\star$ denote a desired interface. We distinguish:

1. *Transient formation*: the trajectory approaches $\Gamma_\star$ within a finite time, possibly with nonzero fluid velocity or acoustic energy.
2. *Driven operation*: constant, periodic, or feedback-controlled actuation maintains acceptable optics and rejects specified disturbances.
3. *Unforced retention*: the shape remains acceptable after the sound, associated flows, and thermal transients have decayed.

A pulse means actuation of finite temporal support, with a specified waveform and bandwidth. It does not imply an ideal Dirac impulse. A periodically repeated pulse train is continued actuation. Solidification is a change of constitutive law and belongs to a distinct retention problem.

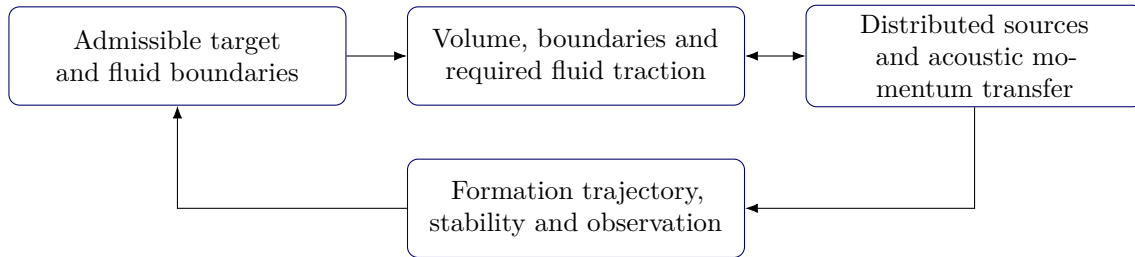


Figure 1: Dependencies of the inverse problem. Changes of shape alter the acoustic field; optical performance must be assessed along the resulting physical state.

1.3 Model hierarchy and mathematical status

The reference continuum description assumes immiscible phases, a regular sharp interface, and no mass transfer through that interface during shaping. A Newtonian stress law is supplied explicitly, while non-Newtonian rheology, solidification, cavitation and topology changes require additional equations. These are physical modeling choices, not assumptions about a particular container. Three conditional reductions are identified where used:

- (i) geometric optics in homogeneous isotropic optical media;
- (ii) small acoustic perturbations with a justified averaging timescale;
- (iii) quiescent, constant-density, constant-surface-tension equilibria for an explicit shape-to-traction formula.

The dynamic formulation retains fluid inertia. Creeping flow appears only as an explicit limiting case. A stationary interface may coexist with streaming; such a state requires a steady-flow solve and is not a quiescent equilibrium.

Derivatives below are conditional on a regular local solution and differentiable state and geometry maps. Smooth interface patches are taken at least C^3 for the displayed curvature variations. We neither prove global well-posedness of the moving-interface problem nor assert existence of a solution for every target. Propositions are stated with the assumptions needed for their proofs. The optimization objectives and design workflow are a proposed synthesis of these

equations. Explicit weak forms, source adjoints and conditional proofs are given for identified subproblems; the full thermoviscous moving-interface system is a reference formulation, not a proved globally solvable inverse. Numerical continuation is neither an assumption nor a step required by the inverse equations. Numerical experiments will test this formulation.

1.4 Notation used throughout

The table defines the principal quantities before their subsequent use. More specialized symbols are introduced immediately before their equations. All physical lengths, times and stresses use SI units unless nondimensionalization is stated.

Symbol	Meaning and units
$\rho_\ell, \mathbf{v}_\ell, p_\ell$	Instantaneous density, velocity and pressure in phase $\ell \in \{-, +\}$; kg m^{-3} , m s^{-1} , Pa.
$\mathbf{T}_\ell, \boldsymbol{\tau}_\ell$	Total Cauchy stress and viscous stress; Pa.
μ_ℓ, ζ_ℓ	Shear and bulk viscosities; Pa s.
$\Theta_\ell, e_\ell, k_\ell$	Temperature, specific internal energy and thermal conductivity; K, J kg^{-1} , $\text{W m}^{-1} \text{K}^{-1}$.
σ, Φ	Surface tension and conservative body-force potential per unit mass; N m^{-1} , $\text{m}^2 \text{s}^{-2}$. The body acceleration is $-\nabla \Phi$.
λ_V	Pressure multiplier imposing lens volume in an incompressible reduction; Pa.
$\mathbf{U}_\ell, p_\ell^h$	Slow hydrodynamic velocity and pressure in an averaged description; m s^{-1} , Pa.
$P_\ell, \mathbf{V}_\ell$	Complex peak acoustic pressure and velocity amplitudes; Pa and m s^{-1} . They are distinct from the slow fields.
$\mathbf{R}_\ell, \mathbf{t}_{\text{ac}}, \Pi$	Acoustic momentum-flux tensor, effective interfacial acoustic traction, and its outward normal component; Pa.
$n_o, n_i, \lambda_{\text{opt}}$	Incident and transmitted optical indices (dimensionless), and vacuum optical wavelength (m).
$\mathbf{O}, \mathbf{I}$	Object and image points; the object subscript is the letter o .
$\eta, \mathbf{w}$	Normal surface displacement (m) and complex peak mechanical emitter velocities (m s^{-1} per component).
$\mathbf{a}, \mathbf{y}$	General actuator command field and full physical state; each may be an infinite-dimensional collection of fields. Units depend on the declared actuator and constitutive model.

Bold symbols denote vectors or tensors. The dagger $\dagger$ denotes conjugate transpose, the superscript T ordinary transpose, and an asterisk complex conjugation. Inner products are real duality pairings unless explicitly complex; Re and Im denote real and imaginary parts.

2 Fluid mechanics and the load required by a target

2.1 A continuum reference problem

For each phase ℓ , define the material derivative $D_t = \partial_t + \mathbf{v}_\ell \cdot \nabla$, the deformation-rate tensor $\mathbf{D}(\mathbf{v}) = (\nabla \mathbf{v} + \nabla \mathbf{v}^T)/2$, and its deviatoric part $\mathbf{D}_0 = \mathbf{D} - (\nabla \cdot \mathbf{v})\mathbf{I}/3$. Let $\mathbf{f}_\ell$ be an additional prescribed body-force density, $\mathbf{q}_\ell$ the conductive heat flux, and r_ℓ an externally supplied volumetric

heating rate. Their units are N m^{-3} , W m^{-2} , and W m^{-3} , respectively. A compressible Newtonian reference model is

$$\partial_t \rho_\ell + \nabla \cdot (\rho_\ell \mathbf{v}_\ell) = 0, \quad (2.1)$$

$$\rho_\ell D_t \mathbf{v}_\ell = \nabla \cdot \mathbf{T}_\ell - \rho_\ell \nabla \Phi + \mathbf{f}_\ell, \quad (2.2)$$

$$\rho_\ell D_t e_\ell = -p_\ell \nabla \cdot \mathbf{v}_\ell + \boldsymbol{\tau}_\ell : \nabla \mathbf{v}_\ell - \nabla \cdot \mathbf{q}_\ell + r_\ell, \quad (2.3)$$

$$\mathbf{T}_\ell = -p_\ell \mathbf{I} + \boldsymbol{\tau}_\ell, \quad \boldsymbol{\tau}_\ell = 2\mu_\ell \mathbf{D}_0(\mathbf{v}_\ell) + \zeta_\ell (\nabla \cdot \mathbf{v}_\ell) \mathbf{I}, \quad \mathbf{q}_\ell = -k_\ell \nabla \Theta_\ell. \quad (2.4)$$

The colon denotes tensor contraction. Equations of state $p_\ell = p_\ell(\rho_\ell, \Theta_\ell)$ and $e_\ell = e_\ell(\rho_\ell, \Theta_\ell)$ close the thermodynamics. Transport coefficients may depend on the local state. Resolving the acoustic waveform in these equations already accounts for its viscous heating and momentum transfer; one must not then add a second radiation-force term for the same wave. Thermoviscous perturbation models derive from these balances, with a consistent separation of orders [4].

Let V_Γ denote interface normal speed. Without mass transfer, $V_\Gamma = \mathbf{v}^- \cdot \mathbf{n} = \mathbf{v}^+ \cdot \mathbf{n}$. The clean-interface stress balance is

$$[[\mathbf{T}]] \mathbf{n} = \sigma \kappa \mathbf{n} - \nabla_\Gamma \sigma. \quad (2.5)$$

For two viscous fluids the usual clean, no-interfacial-slip model also imposes velocity continuity. Other interfacial constitutive laws must be specified if surface viscosity, surfactants, or slip are admitted. With constant surface tension, a static spherical inclusion obeys $p^- - p^+ = 2\sigma/R_s$, where R_s is its radius; this checks the sign in [equation \(2.5\)](#).

A thermally passive interface with no thermal resistance or surface heat storage has continuous temperature and normal conductive heat flux. If σ varies with temperature or composition, the associated surface thermodynamics and species transport must accompany the Marangoni term in [equation \(2.5\)](#). Keeping that force while claiming an exact constant- σ energy identity would be inconsistent.

The remaining boundaries require mechanical and thermal conditions: for example prescribed motion, an elastic solid coupled by traction and velocity, or a radiating exterior. These alternatives define different wave and flow operators. Neither a rigid reflecting chamber nor a pressure-release free surface is a universal boundary condition.

2.2 Wetting is part of the problem data

If a supporting solid is present, let θ denote contact angle measured through the lens fluid, and let γ_{s-} and γ_{s+} be the solid's interfacial free energies with the two phases, in N m^{-1} . An ideal equilibrium contact angle θ_e satisfies Young's relation

$$\sigma \cos \theta_e = \gamma_{s+} - \gamma_{s-}. \quad (2.6)$$

A fixed contact line is a positional constraint; it is not equivalent to prescribing this angle. For a mobile line, its law must be supplied. As one explicit dissipative example, define V_C positive when the lens fluid advances over the wall, and a line-friction coefficient $\zeta_C \geq 0$ in Pa.s. Then

$$\zeta_C V_C = \sigma (\cos \theta_e - \cos \theta) \quad (2.7)$$

is a possible closure, not a universal wetting law. Hysteresis can instead allow a stationary line over a measured interval of angles. The classical moving-line/no-slip singularity means that imposing motion also requires a compatible microscopic regularization, such as slip or a diffuse-interface model [5, 6]. Neither a guessed contact angle nor a visually retained meniscus replaces these data.

2.3 Variational derivation of the required static traction

Now restrict attention to constant-density fluids, constant σ , a fixed support, and a quiescent state with no body force beyond $-\rho_\ell \nabla \Phi$. The varied lens region has finite volume and deformable interfacial area. Set $\Delta\rho = \rho_- - \rho_+$. Let $A[\Gamma]$ denote deformable interfacial area, $V[\Gamma]$ the lens volume, and E_{wall} the solid wetting energy. Up to constants independent of the shape, the mechanical potential energy is

$$E[\Gamma] = \sigma A[\Gamma] + \Delta\rho \int_{\Omega^-} \Phi \, dV + E_{\text{wall}}. \quad (2.8)$$

This is the standard capillary variational structure [7]. For a virtual displacement $\eta \mathbf{n}$ compatible with the support, area and volume variations give

$$\delta V = \int_{\Gamma} \eta \, dA, \quad \delta E = \int_{\Gamma} (\sigma \kappa + \Delta\rho \Phi) \eta \, dA + \text{contact-line terms}. \quad (2.9)$$

The line terms either vanish for a pinned boundary or supply the appropriate wetting condition. A normal acoustic traction Π , positive outward, does virtual work $\int_{\Gamma} \Pi \eta \, dA$. Enforcing the prescribed volume V_0 with the multiplier λ_V therefore gives

$$\boxed{\Pi_{\text{req}}[\Gamma] = \sigma \kappa + \Delta\rho \Phi - \lambda_V, \quad V[\Gamma] = V_0.} \quad (2.10)$$

Here V_0 is a supplied volume, not a fitted correction to the target. Every term in the traction equation is a pressure. It determines the load required by any admissible smooth target, whether Cartesian or otherwise.

For a graph $z = h(x, y)$ with the lens fluid below it and an upward normal, let ∇_{xy} act in the horizontal plane. In a uniform downward gravitational field with magnitude $g_{\text{grav}} > 0$, $\Phi = g_{\text{grav}} z$, and

$$\kappa = -\nabla_{xy} \cdot \frac{\nabla_{xy} h}{\sqrt{1 + |\nabla_{xy} h|^2}}, \quad \Pi_{\text{req}} = -\sigma \nabla_{xy} \cdot \frac{\nabla_{xy} h}{\sqrt{1 + |\nabla_{xy} h|^2}} + \Delta\rho g_{\text{grav}} h - \lambda_V. \quad (2.11)$$

No small-slope approximation is made. Overhangs require a surface representation, using [equation \(2.10\)](#) directly.

Define the zero-mean projection of a surface scalar f by $\mathcal{P}_0 f = f - A[\Gamma]^{-1} \int_{\Gamma} f \, dA$. For closed-volume shape variations, only the nonconstant part of the normal load determines shape:

$$\mathcal{P}_0 \Pi = \mathcal{P}_0 (\sigma \kappa + \Delta\rho \Phi). \quad (2.12)$$

The omitted constant is balanced by the pressure multiplier. This gauge does not remove restrictions on absolute pressure, cavitation, or actuator loading. For several independently conserved volumes there is one corresponding constraint per volume. A reservoir changes this formulation according to its specified pressure and mass exchange.

The quiescent reduction additionally requires compatible tangential traction and no uncompensated mean bulk forcing. If acoustic streaming produces a stationary flow $\mathbf{U}_\star \neq 0$, its pressure and viscous stresses must be solved together with $\mathbf{U}_\star \cdot \mathbf{n} = 0$. Matching [equation \(2.10\)](#) alone does not establish that steady state.

2.4 What a finite pulse can leave behind

Proposition 2.1 (Necessary condition for unforced retention). *Suppose actuation ends, all acoustic, flow and thermal transients decay, the liquid remains liquid with the same constant*

densities and surface tension, and the support, wetting law, volume and conservative body force are unchanged. Any smooth quiescent limiting interface must satisfy

$$\sigma\kappa + \Delta\rho\Phi = \lambda_V \quad (2.13)$$

with its volume and boundary conditions. A target violating this condition cannot be a permanently retained quiescent result of that pulse.

Proof. At a quiescent limit, hydrostatic balance gives $p_-^h - p_+^h = \lambda_V - \Delta\rho\Phi$. With no residual wave traction, [equation \(2.5\)](#) requires $p_-^h - p_+^h = \sigma\kappa$. Equivalently, the constrained first variation of [equation \(2.8\)](#) vanishes. Both give [equation \(2.13\)](#). $\square$

This is a necessary condition for a limit, not a proof that every trajectory converges or that equilibrium is unique. Persistent rotation or another non-quiescent relative equilibrium requires retaining that flow and its conserved quantities; it is outside the stated quiescent-limit hypothesis. Contact-angle hysteresis can leave a different admissible pinned boundary. Nevertheless, it cannot support an arbitrary non-capillary interior shape in a fully relaxed liquid. Continuous actuation, a modified support, or curing can alter the retention question. Fluidic fabrication by shaping and solidification is an experimentally demonstrated but distinct route [8]; it is not evidence that an arbitrary shape persists after an acoustic pulse in an unchanged liquid.

3 Geometric constraints, dual traction, and two lens faces

The organizing inverse map is

$$\text{target geometry} \longrightarrow \text{required traction} \longrightarrow \text{continuous sources} \longrightarrow \text{finite realization.}$$

Formation and stability test the resulting sources in the forward dynamics. Every arrow after the geometric calculation is a feasibility problem, not an unconditional implication of success. Cartesian geometry will specialize this construction in [section 16](#); the equations apply to arbitrary admissible interfaces and three-dimensional perturbations.

3.1 The geometric multiplier is a traction field

For a graph with the lens below it, let D be its horizontal domain and $D_o \subset D$ the prescribed optical aperture. With pinned support, constant surface tension, and uniform gravity, write

$$E_0[h] = \int_D \left(\sigma \sqrt{1 + |\nabla h|^2} + \frac{1}{2} \Delta\rho g_{\text{grav}} h^2 \right) dx dy, \quad (3.1)$$

$$\mathcal{L} = E_0[h] - \lambda_V (V[h] - V_0) - \int_{D_o} \Lambda(\mathbf{x})(h - h_\star) dx dy. \quad (3.2)$$

Here Λ has pressure units. Its multiplier term is virtual work because normal displacement times surface area equals vertical displacement times horizontal area. A scalar cannot enforce the pointwise constraint.

Proposition 3.1 (Dual field of prescribed geometry). *For a regular compatible stationary point of [equation \(3.2\)](#), the geometric multiplier on D_o is*

$$\Lambda = -\sigma \nabla \cdot \frac{\nabla h_\star}{\sqrt{1 + |\nabla h_\star|^2}} + \Delta\rho g_{\text{grav}} h_\star - \lambda_V = \Pi_{\text{req}}. \quad (3.3)$$

Proof. Variation with respect to Λ enforces the target, variation with respect to λ_V enforces volume, and integration by parts in the h variation gives [equation \(3.3\)](#). Boundary terms vanish for admissible pinned variations or give the specified natural support conditions. $\square$

For a fully prescribed compatible graph, its volume constraint is redundant; the resulting dependence of the constraints explains the additive gauge in Λ . Once the shape is known, this is an explicit differential operator on the shape, not a separate elliptic PDE that determines acoustic pressure. The mean traction Π and the oscillatory acoustic pressure phasor P have the same units but different physical meanings.

With a shape-independent load, the second variation is

$$\delta^2 E_0[\eta] = \int_D \left[\sigma \frac{(1 + |\nabla h|^2)|\nabla \eta|^2 - (\nabla h \cdot \nabla \eta)^2}{(1 + |\nabla h|^2)^{3/2}} + \Delta \rho g_{\text{grav}} \eta^2 \right] dx dy. \quad (3.4)$$

For $\Delta \rho \geq 0$, positive σ , and a nonempty pinned boundary, this is positive for nonzero admissible variations. Thus a compatible critical graph is the unique minimizer within this convex graph domain under an ideal prescribed load. Acoustic actuation has a shape-dependent load, whose derivative must still be subtracted in [equation \(11.5\)](#).

3.2 Front and back in one apparatus

Let the lens occupy $h_f(x, y) < z < h_b(x, y)$, between front and back surfaces. The outward normals point downward on the front and upward on the back. In a common external fluid of density ρ_e , set $\Delta \rho = \rho_L - \rho_e$. Ignoring constants from fixed supports,

$$E_0 = \int_D \left[\sigma_f \sqrt{1 + |\nabla h_f|^2} + \sigma_b \sqrt{1 + |\nabla h_b|^2} + \frac{1}{2} \Delta \rho g_{\text{grav}} (h_b^2 - h_f^2) \right] dx dy, \quad (3.5)$$

$$V_L = \int_D (h_b - h_f) dx dy. \quad (3.6)$$

The acoustic work is $\int_D (\Pi_b \delta h_b - \Pi_f \delta h_f) dx dy$. Consequently the outward required loads are

$$\Pi_f = +\sigma_f \nabla \cdot \frac{\nabla h_f}{\sqrt{1 + |\nabla h_f|^2}} + \Delta \rho g_{\text{grav}} h_f - \lambda_V, \quad (3.7)$$

$$\Pi_b = -\sigma_b \nabla \cdot \frac{\nabla h_b}{\sqrt{1 + |\nabla h_b|^2}} + \Delta \rho g_{\text{grav}} h_b - \lambda_V. \quad (3.8)$$

There is one shared lens pressure multiplier when the exterior is connected and hydrostatically equilibrated. Subtracting a separate mean from each face would discard the relative load that sets thickness. The pressure gauge is the constant function on the *union* of the faces, not two unrelated constants. The off-diagonal derivatives $D_{h_b} \Pi_f$ and $D_{h_f} \Pi_b$ are generally nonzero because both faces scatter the same wave.

A sealed three-layer cylinder is a different, useful test apparatus. Let ρ_0, ρ_1, ρ_2 denote the lower, lens, and upper fluid densities. For upward graph coordinates, set $d_f = \rho_0 - \rho_1$ and $d_b = \rho_1 - \rho_2$. Its gravitational energy is $\int_D g_{\text{grav}} (d_f h_f^2 + d_b h_b^2) / 2 dx dy$. It separately conserves lower-fluid volume and lens volume; upper-fluid volume then follows from the total cylinder volume. In equivalent coordinates, both integrals $\int_D h_f$ and $\int_D h_b$ are fixed. The two associated pressure differences are legitimate constraint multipliers. Required upward loads are

$$q_j = -\sigma_j \nabla \cdot \frac{\nabla h_j}{\sqrt{1 + |\nabla h_j|^2}} + d_j g_{\text{grav}} h_j - \lambda_j, \quad j \in \{f, b\}. \quad (3.9)$$

Here $q_f = -\Pi_f$ and $q_b = \Pi_b$. Stable stratification $d_f, d_b \geq 0$ makes the passive graph energy convex. A dense lens in one connected lighter bath does not have that same two-face convexity; the lower face carries the opposite gravitational sign. Reservoirs, bypass channels, and moving supports must be specified rather than introduced through fitted pressure constants.

3.3 The non-optical annulus is a design variable

Prescribe $h_j = h_{C,j}$ only on each clear aperture. Outside it minimize a declared functional, such as E_0 or the area-weighted squared required load, subject to all volume constraints, support values, clearance and regularity. Energy minimization does not minimize source power; that needs the acoustic map. A graph with a slope jump at the aperture edge requires a concentrated line force. A classical bounded-traction design therefore needs slope matching; continuous traction usually demands curvature matching as well. For a load-norm objective, an explicit C^2 or Sobolev regularity class and its matching constraints prevent hidden line actuators. The optimum annulus is coupled to the common wave solution on both faces.

4 Acoustic transfer, radiation traction and temporal memory

4.1 When a harmonic reduction is justified

Let $\omega > 0$ be angular carrier frequency, $T_a = 2\pi/\omega$ its period, T_e an envelope variation time, T_Γ an interface variation time, and τ_r an acoustic energy ring-down time. A fast-time average over an interval Δt requires

$$T_a \ll \Delta t \ll \min(T_e, T_\Gamma). \quad (4.1)$$

Replacing the field at each instant by a settled harmonic solution also requires $\tau_r \ll \min(T_e, T_\Gamma)$. A short carrier period alone does not imply short ring-down in a resonant cavity. When these conditions fail, causal wave dynamics or the reference continuum equations must be retained.

For a fixed slow state, use the peak-phasor convention $p_{1\ell}(\mathbf{x}, t) = \text{Re}\{P_\ell(\mathbf{x})e^{-i\omega t}\}$ and $\mathbf{v}_{1\ell}(\mathbf{x}, t) = \text{Re}\{\mathbf{V}_\ell(\mathbf{x})e^{-i\omega t}\}$, where $i^2 = -1$ and the subscript 1 denotes first-order acoustics. Thus the average of two real harmonic quantities with phasors A, B is $\frac{1}{2} \text{Re}(AB^*)$. For locally homogeneous, inviscid, adiabatic fluids with negligible mean advection, define the sound speed c_ℓ and obtain

$$-i\omega\rho_\ell\mathbf{V}_\ell = -\nabla P_\ell, \quad -i\omega P_\ell = -\rho_\ell c_\ell^2 \nabla \cdot \mathbf{V}_\ell, \quad \nabla \cdot \left(\frac{\nabla P_\ell}{\rho_\ell} \right) + \frac{\omega^2}{\rho_\ell c_\ell^2} P_\ell = 0. \quad (4.2)$$

The sound speed is the isentropic derivative of the equation of state. Loss, thermal fields and elastic boundaries replace this simple operator when needed; they are not unspecified tuning coefficients. Boundary-layer reductions with declared scale conditions are developed in [9, 4].

The acoustic interface conditions must follow by linearizing the kinematic and traction conditions on the displaced interface. Let Ξ be its first-order normal-displacement phasor, in metres. About a quiescent interface, $\mathbf{V}^- \cdot \mathbf{n} = \mathbf{V}^+ \cdot \mathbf{n} = -i\omega\Xi$. First-order traction also contains the capillary curvature variation and the displacement through base pressure gradients. Neglecting those corrections can yield continuity of acoustic pressure and normal velocity, as in a common high-frequency fluid–fluid scattering model [2]. That simplification requires the fast interfacial restoring terms to be small; it is not an exact condition for capillary waves at all frequencies. A compliant solid or a free surface requires its own linearized boundary operator.

4.2 The sign and content of radiation traction

For the locally homogeneous inviscid acoustic reduction just specified, define the time-averaged acoustic *momentum flux*

$$\mathbf{R}_\ell = \left(\frac{|P_\ell|^2}{4\rho_\ell c_\ell^2} - \frac{\rho_\ell |\mathbf{V}_\ell|^2}{4} \right) \mathbf{I} + \frac{\rho_\ell}{2} \text{Re}(\mathbf{V}_\ell \otimes \mathbf{V}_\ell^*). \quad (4.3)$$

It is minus the radiation-stress convention used in [2]; recording this sign is essential. The effective outward interfacial traction and its scalar normal component are

$$\mathbf{t}_{\text{ac}} = (\mathbf{R}_- - \mathbf{R}_+)\mathbf{n} = -\llbracket \mathbf{R} \rrbracket \mathbf{n}, \quad \Pi = \mathbf{n} \cdot \mathbf{t}_{\text{ac}}. \quad (4.4)$$

Let $\mathbf{T}^h = -p^h \mathbf{I} + \boldsymbol{\tau}(\mathbf{U})$ denote slow hydrodynamic stress with the mean-pressure convention associated with equation (4.3). The reduced interface balance becomes

$$\llbracket \mathbf{T}^h \rrbracket \mathbf{n} + \mathbf{t}_{\text{ac}} = \sigma \kappa \mathbf{n} - \nabla_{\Gamma} \sigma. \quad (4.5)$$

In a consistent bulk momentum balance the acoustic force is $-\nabla \cdot \mathbf{R}$ with the matching mean-pressure definition. A gradient part can be absorbed into p^h ; it must not be counted again at the interface. In thermoviscous or inhomogeneous fluids, the averaged stress, heat source and boundary-layer forcing have to be derived together from the reference equations. A scalar normal pressure model does not cover all streaming or tangential stresses.

Only in a one-sided pressure-release limit, with negligible adjoining-fluid momentum flux, $P_- \simeq 0$ and negligible tangential acoustic velocity at the surface, does equation (4.4) reduce to

$$\Pi \simeq \frac{\rho_-}{4} |\mathbf{V}_- \cdot \mathbf{n}|^2. \quad (4.6)$$

This factor uses peak amplitudes. It must not be transferred unchanged to an immersed liquid–liquid lens or to a surface supporting appreciable acoustic pressure. Experiments on liquid interfaces and coupled calculations motivate retaining the wave–shape interaction, rather than prescribing a load independent of the deformed geometry [1, 2, 3].

4.3 Impulses are causal histories, not instantaneous intensities

For a fixed geometry and linear acoustics, let $\mathbf{z}_a(t)$ collect the wave state, $\mathcal{G}_{\Gamma}(t)$ its causal propagation operator, and $\mathcal{B}_{\Gamma}$ the command-to-source operator. Let t_0 be the starting time and $\mathbf{z}_{a,0}$ the initial acoustic state. Then

$$\mathbf{z}_a(t) = \mathcal{G}_{\Gamma}(t - t_0) \mathbf{z}_{a,0} + \int_{t_0}^t \mathcal{G}_{\Gamma}(t - s) \mathcal{B}_{\Gamma} \mathbf{a}(s) \, ds. \quad (4.7)$$

The radiative load is quadratic in this accumulated field. Two commands with the same present amplitude can therefore yield different loads because their histories differ. Changing the interface makes the operator time dependent, so a fixed convolution kernel is then only a local approximation. Fluid inertia, viscous relaxation, streaming, temperature and actuator mechanics provide additional memory variables.

An actual impulse has a restricted mechanical interpretation. Consider a declared inertial modal reduction with displacement coordinates $\mathbf{q}_f$, modal mass matrix $\mathbf{M}_f$ and generalized acoustic force $\mathbf{f}_a(t)$. Their component units are metres, kilograms and newtons, respectively. Let t_-, t_+ bracket the effective force pulse, including any relevant acoustic ring-down. If the shape and $\mathbf{M}_f$ change negligibly over that interval and integrated restoring and dissipative forces are small, integration of momentum gives

$$\mathbf{M}_f [\dot{\mathbf{q}}_f(t_+) - \dot{\mathbf{q}}_f(t_-)] \simeq \int_{t_-}^{t_+} \mathbf{f}_a(t) \, dt. \quad (4.8)$$

The integral has units of momentum. In this limit a short force pulse primarily changes velocity; the subsequent interface shape follows the inertial, capillary and viscous evolution. This reduction does not apply when viscous forces dominate the pulse interval, and an acoustic cycle average still needs equation (4.1). Delivering the correct integrated force is not the same as arriving at the target with negligible residual motion.

If several carriers ω_m are used, index m labels their frequencies. Quadratic products contain cross terms proportional to $e^{-i(\omega_m - \omega_n)t}$. They average away only with appropriate coherence statistics or an averaging interval satisfying $|\omega_m - \omega_n|\Delta t \gg 1$, together with negligible relevant mechanical response to the unresolved beats. Nearby carriers can drive slow beat forces; a sum of independent intensities is not generally their physical load.

5 Distributed sources and their forward operator

5.1 Source distributions precede an array

Fix a regular interface, material state and frequency satisfying the harmonic assumptions of [section 4](#). Let $\Gamma_a \subseteq \partial\mathcal{D}_c$ be the permitted boundary source support, and let $D_s \subset \mathcal{D}_c$ be the permitted volume source support. Either can have several components and can be empty. Define the complex peak source fields

$$u = (g, \mathbf{F}, q), \quad [g] = \text{m s}^{-1}, \quad [\mathbf{F}] = \text{N m}^{-3}, \quad [q] = \text{s}^{-1}. \quad (5.1)$$

Here g is imposed outward boundary velocity, $\mathbf{F}$ is a first-order body-force density, and q is an optional volumetric dilation rate in the linear acoustic equations. Volume fields are understood phase by phase and vanish outside D_s . No restriction to radial, annular, or equal-phase sources is made. Their physical time signals are the real parts of their phasors multiplied by $e^{-i\omega t}$.

The dilation source is a generalized acoustic representation, not an extra unexplained mass supply in [equation \(2.1\)](#). An implementation must derive it from its mechanism, such as a moving source volume or heat input, and supply the associated mean mass, energy and thermal balances. In a closed, mass-conserving mechanical realization one can set $q = 0$ and use boundary motion and body forcing. An acoustic source description alone does not specify its mean mechanical reaction or heat production.

For definiteness choose positive reference area S_a , source volume V_s , and amplitudes g_0, F_0, q_0 . Define a complex Hilbert source space $\mathcal{U}$ with squared norm

$$\|u\|_{\mathcal{U}}^2 = \frac{1}{S_a g_0^2} \int_{\Gamma_a} |g|^2 dA + \frac{1}{V_s F_0^2} \int_{D_s} |\mathbf{F}|^2 dV + \frac{1}{V_s q_0^2} \int_{D_s} |q|^2 dV. \quad (5.2)$$

Omit components that are not available. This norm is dimensionless; it is not automatically power. The corresponding complex inner product is conjugate linear in its first argument. Spatial smoothness can instead be penalized with a Sobolev norm. The admissible subset $\mathcal{U}_{\text{ad}}$ specifies support, amplitude, smoothness and any hardware restrictions. Temporal bandwidth is a condition on a waveform or envelope, not on a single phasor.

The displayed weak and trace equations assume regular enough source fields and solutions. In the explicit derivation, $\mathbf{F}, q$ are supported away from the interface and outer wall. Separated source and observation sets permit additional interior regularity; singular point sources need a different function-space treatment. Measure-valued Helmholtz source inversion is one such treatment [\[10\]](#); its existence and sparse-recovery results do not automatically apply to a quadratic interface-stress objective.

5.2 An explicit transmission problem

The following is a useful, limited member of the general model: constant real ρ_ℓ, c_ℓ in each phase, negligible first-order capillary compliance, negligible mean advection, and inviscid bulk acoustics. Dissipation is represented here by a declared boundary admittance. The forced equations are

$$-i\omega\rho_\ell\mathbf{V}_\ell = -\nabla P_\ell + \mathbf{F}_\ell, \quad -\frac{i\omega}{\rho_\ell c_\ell^2}P_\ell = -\nabla \cdot \mathbf{V}_\ell + q_\ell. \quad (5.3)$$

Define $\alpha_\ell = 1/\rho_\ell$ and $\beta_\ell = \omega^2/(\rho_\ell c_\ell^2)$. Eliminating velocity gives

$$L_\ell P_\ell := -\nabla \cdot (\alpha_\ell \nabla P_\ell) - \beta_\ell P_\ell = -\nabla \cdot (\alpha_\ell \mathbf{F}_\ell) - i\omega q_\ell. \quad (5.4)$$

The colon-equals sign defines the operator L_ℓ . With ∂_n taken along the same minus-to-plus normal on both sides, transmission is

$$[[P]] = 0, \quad [[\alpha \partial_n P]] = 0, \quad v = \frac{\alpha_\ell \partial_n P_\ell}{i\omega}. \quad (5.5)$$

The scalar v is the common acoustic normal-velocity trace; its units are velocity. Pressure continuity here is the stated high-frequency reduction, not a replacement for the full interface conditions at all frequencies.

Let $\mathbf{n}_a$ be the domain-outward normal on the cylinder boundary, and let $Y(\mathbf{x}, \omega)$ be a prescribed acoustic admittance with units $\text{m Pa}^{-1} \text{s}^{-1}$. Set $g = 0$ outside Γ_a and impose

$$\mathbf{V} \cdot \mathbf{n}_a = YP + g, \quad \alpha \partial_{n_a} P = i\omega(YP + g). \quad (5.6)$$

The second equation uses the stated absence of $\mathbf{F}$ near the wall. The case $Y = 0$ is prescribed wall velocity. A passive local admittance has $\text{Re } Y \geq 0$; arbitrary elastic wall dynamics need a coupled operator in place of this scalar relation. An open exterior also needs its own radiation condition. Even with passive boundaries, well-posedness at the chosen frequency must be established; an undamped cavity at an eigenfrequency is not covered by the inverse operator used below.

For a complex, globally continuous test pressure $\psi \in H^1(\mathcal{D}_c)$, where H^1 means a square-integrable function with square-integrable first derivatives, integration by parts yields $a_\Gamma(P, \psi) = b(u, \psi)$, with

$$a_\Gamma(P, \psi) = \sum_\ell \int_{\Omega_\ell} (\alpha_\ell \nabla P_\ell \cdot \nabla \bar{\psi} - \beta_\ell P_\ell \bar{\psi}) dV - i\omega \int_{\partial \mathcal{D}_c} YP \bar{\psi} dA, \quad (5.7)$$

$$b(u, \psi) = i\omega \int_{\Gamma_a} g \bar{\psi} dA + \int_{D_s} \alpha \mathbf{F} \cdot \nabla \bar{\psi} dV - i\omega \int_{D_s} q \bar{\psi} dV. \quad (5.8)$$

A bar denotes complex conjugation. The weak form defines exactly how the distributed controls enter the wave operator. It can be replaced systematically by a thermoviscous or solid-coupled weak form; those replacements also change the load and its adjoint. Established active Helmholtz control illustrates the importance of source support, resonance and regularization [11]. That literature controls compatible wave fields in stated regions; it is not a theorem of arbitrary fluid-shape controllability.

5.3 Input power is an independent consistency condition

Define the net time-averaged acoustic source power entering this canonical problem by

$$P_{\text{src}} = -\frac{1}{2} \text{Re} \int_{\Gamma_a} P \bar{g} dA + \frac{1}{2} \text{Re} \int_{D_s} \mathbf{F} \cdot \bar{\mathbf{V}} dV + \frac{1}{2} \text{Re} \int_{D_s} P \bar{q} dV. \quad (5.9)$$

The minus sign for g follows from its outward convention: positive work into the fluid uses traction $-P\mathbf{n}_a$. Every term has units of watts. In a settled state with lossless bulk coefficients, the power identity is

$$P_{\text{src}} = \frac{1}{2} \int_{\partial \mathcal{D}_c} \text{Re } Y |P|^2 dA. \quad (5.10)$$

To derive it, put $\psi = P$ in equations (5.7) and (5.8), take $-\text{Im}/(2\omega)$, and substitute $\mathbf{V} = \alpha(\nabla P - \mathbf{F})/(i\omega)$. The term $\mathbf{F} \cdot \bar{\mathbf{F}}$ contributes no real source work in that substitution. Real bulk loss, mean work, or unresolved wall layers require their corresponding terms in a more complete power balance. Net acoustic power is distinct from gross electrical consumption and from equation (5.2).

5.4 Cylinder symmetry does not impose source symmetry

Let (r, θ, z) be cylindrical coordinates, with $r^2 = x_1^2 + x_2^2$. Only when geometry, material state and boundary operators are invariant under rotation can the linear wave problem be separated into integer azimuthal orders $m \in \mathbb{Z}$, proportional to $e^{im\theta}$. Products of an order- m field with the conjugate of an order- n field have order $m - n$. Thus quadratic radiation loads contain differences of the excited angular orders. One pure angular order produces an axisymmetric mean normal load in this invariant configuration; coherent combinations can produce non-axisymmetric loads. Axisymmetric source fields alone cannot produce a non-axisymmetric load in this fixed, axisymmetric linear wave problem. Once the interface or boundary breaks that symmetry, angular orders couple. This selection rule follows from the displayed product, and is a testable consequence of the source formulation rather than an array-design assumption.

6 Local trace synthesis and its geometric cost

The geometric multiplier gives a target load, not independent acoustic Cauchy data. Nevertheless, a conditional construction clarifies what very rich source access would permit. This section is a deduction for the inviscid transmission reduction; it is not an experimental fabrication result.

6.1 Two-fluid traces and a pressure node

Pressure and normal velocity are continuous at the ideal interface. Write their traces as P and v , and $\Delta\rho = \rho_- - \rho_+$. Eliminating tangential velocities from [equation \(4.3\)](#) gives

$$\Pi = \frac{1}{4} \underbrace{\left(\frac{1}{\rho_- c_-^2} - \frac{1}{\rho_+ c_+^2} \right)}_{c_P} |P|^2 + \frac{\Delta\rho}{4} |v|^2 + \frac{\Delta\rho}{4\omega^2 \rho_- \rho_+} |\nabla_\Gamma P|^2. \quad (6.1)$$

The tangential radiation-traction jump vanishes in this ideal model. The coefficients and orientation follow from the momentum-flux convention, as in the coupled-wave approach of [2].

Set $H = \sigma\kappa + \Delta\rho\Phi$. If $\Delta\rho \neq 0$ and an admissible pressure offset makes $(H - \lambda_V)/\Delta\rho > 0$, then

$$P|_\Gamma = 0, \quad v = 2\sqrt{(H - \lambda_V)/\Delta\rho} \quad (6.2)$$

realizes the required normal load at the trace level. If $\Delta\rho = 0$ but $c_P \neq 0$, the alternative $v = 0$, $P = \sqrt{(H - \lambda_V)/c_P}$ does so. Equal density and sound speed give zero ideal radiation-traction contrast, irrespective of optical-index contrast. These trace prescriptions have yet to be produced by admissible sources.

6.2 Conditional local fields and volume forces

For analytic target patches and analytic positive radicand, let φ_ℓ solve $(\Delta + k_\ell^2)\varphi_\ell = 0$ locally with $\varphi_\ell = 0$, $\partial_n \varphi_\ell = v$ on the patch and $k_\ell = \omega/c_\ell$. The analytic Cauchy theorem [12] applies to this noncharacteristic surface. It is a local existence statement; elliptic Cauchy continuation to remote walls need not be stable. Let smooth cutoffs χ_ℓ equal one near the patch and vanish inside the available analytic neighborhood. With $\tilde{\varphi}_\ell = \chi_\ell \varphi_\ell$, define

$$\mathbf{V}_\ell = \nabla \tilde{\varphi}_\ell, \quad P_\ell = \frac{\rho_\ell c_\ell^2}{i\omega} \Delta \tilde{\varphi}_\ell, \quad (6.3)$$

$$\mathbf{F}_\ell = \frac{\rho_\ell c_\ell^2}{i\omega} \nabla (\Delta + k_\ell^2) \tilde{\varphi}_\ell, \quad q_\ell = 0. \quad (6.4)$$

Substitution verifies both forced first-order acoustic balances. The force is supported where the cutoff changes, leaving the prescribed interface traces unaltered. This requires independently realizable volume forces on both sides, with compatible mean reactions and energy budgets. At contact lines and patch edges the construction requires additional matching. For two separated faces, disjoint source neighborhoods may supply local constructions, but a single boundary-only Helmholtz problem imposes additional joint compatibility. Neither local construction proves controllability from a cylindrical array.

6.3 A necessary bound for strict geometric accuracy

Let ϵ_a bound peak normal carrier displacement. Since $v = -i\omega\Xi$, a pressure-node realization obeys $|\Pi| \leq |\Delta\rho|\omega^2\epsilon_a^2/4$. Its load has one sign. Consequently a necessary condition, independent of the permitted pressure offset, is

$$\text{osc}_\Gamma H \leq \frac{|\Delta\rho|\omega^2\epsilon_a^2}{4}, \quad f \geq \frac{1}{\pi\epsilon_a} \sqrt{\frac{\text{osc}_\Gamma H}{|\Delta\rho|}}. \quad (6.5)$$

Here $\text{osc } H = \sup H - \inf H$. A strictly positive load margin increases this cost. For graph-height error the vertical oscillation at fixed horizontal position is Ξ/n_z , so the pointwise height bound is stronger where $|n_z| < 1$. This inequality rejects a particular realization when violated; pressure-dominated or tangential-field realizations may avoid it. Raising frequency also changes attenuation, thermal loads, acoustic resolution and actuator response, so it is not by itself a demonstrated solution.

7 Attainable loads and a regularized inverse

7.1 Continuous quadratic load operators

At fixed geometry let $\mathcal{A}_\Gamma$ denote a well-posed linear acoustic operator including its boundary and interface conditions, and let $\mathcal{B}_\Gamma$ map a source field to its right-hand side. The acoustic state $\mathbf{z}_a$ satisfies

$$\mathcal{A}_\Gamma \mathbf{z}_a = \mathcal{B}_\Gamma u. \quad (7.1)$$

Let $\mathcal{S}_\Gamma : \mathcal{U} \rightarrow \mathcal{X}$ map sources to the pressure and velocity traces needed for the radiation traction. Here $\mathcal{X}$ is a Hilbert trace space, with fixed physical scaling, continuously embedded in the product of $L^4(\Gamma)$ spaces for those components. The latter space contains functions with integrable fourth power. This regularity ensures that quadratic loads lie in $L^2(\Gamma)$. Such trace regularity is an assumption on this reduction, not a property of every sharp-edged or singular-source configuration.

For any real dimensionless test function ϕ compatible with the surface support and with $\int_\Gamma \phi \, dA = 0$, define the generalized force and the required force by

$$F_\phi(u) = \int_\Gamma \phi \Pi(u) \, dA, \quad b_\phi = \int_\Gamma \phi(\sigma\kappa + \Delta\rho\Phi) \, dA. \quad (7.2)$$

Both have units of newtons. Bounded tests suffice; more general tests are allowed where the pairing is continuous. The radiation expression defines a bounded Hermitian quadratic form $\mathcal{C}_\phi$ on $\mathcal{X}$. With the star denoting the Hilbert adjoint in this subsection, set

$$\mathcal{K}_\phi = \mathcal{S}_\Gamma^* \mathcal{C}_\phi \mathcal{S}_\Gamma, \quad \boxed{F_\phi(u) = \langle u, \mathcal{K}_\phi u \rangle_{\mathcal{U}} = b_\phi.} \quad (7.3)$$

The operator $\mathcal{K}_\phi$ is self-adjoint, although it need not be positive. Because the source norm is dimensionless, its quadratic form has force units. Exact quiescent normal balance requires this

equality for a dense set of admissible tests, with the wetting and volume constraints. Tangential balance and absence of incompatible mean bulk forcing remain separate requirements. A steady streaming state instead uses the full stationary equations.

At fixed geometry the first variation is

$$DF_\phi(u)[\delta u] = 2 \operatorname{Re} \langle \mathcal{K}_\phi u, \delta u \rangle_{\mathcal{U}}. \quad (7.4)$$

Multiplication of all sources by the same phase $e^{i\vartheta}$, for real ϑ , leaves the load unchanged. Consequently $\delta u = iu \delta\vartheta$ is a null direction of this derivative. Sources invisible to the relevant wave traces add further nonuniqueness. At $u = 0$ the derivative vanishes; this does not exclude finite-amplitude shaping. For infinitesimal target changes, the differentiated required load, including the change of $\mathcal{K}_\phi$ with geometry, must lie in the admissible derivative range. This is a condition of local authority, not a prescription to generate the family by continuation.

7.2 Necessary source-effort and sign bounds

Choose a finite set of tests ϕ_i , real coefficients a_i , and write $b_i = b_{\phi_i}$, $\mathcal{K}_i = \mathcal{K}_{\phi_i}$ and $\mathcal{K}_a = \sum_i a_i \mathcal{K}_i$. Let $C_u > 0$ be an upper bound on $\|u\|_{\mathcal{U}}$ and let $\|\mathcal{K}_a\|$ denote operator norm. Every exact source under that bound satisfies

$$\left| \sum_i a_i b_i \right| \leq C_u^2 \|\mathcal{K}_a\|, \quad \mathcal{K}_a \succeq 0 \implies \sum_i a_i b_i \geq 0. \quad (7.5)$$

Here $\succeq 0$ means a nonnegative quadratic form. The inequalities follow directly from [equation \(7.3\)](#) and the operator-norm bound. If $\|\mathcal{K}_a\| > 0$, they also give the lower bound $\|u\|_{\mathcal{U}}^2 \geq |\sum_i a_i b_i| / \|\mathcal{K}_a\|$. If the operator is zero but the target combination is nonzero, exact matching is impossible in this model. These are certificates of necessary conditions; passing them does not establish reachability, stability or uniqueness.

A power version needs an additional hypothesis. Suppose input power is the quadratic form $\langle u, \mathcal{R}_P u \rangle_{\mathcal{U}}$ with a bounded, coercive self-adjoint operator $\mathcal{R}_P$, meaning $\langle u, \mathcal{R}_P u \rangle \geq r_P \|u\|_{\mathcal{U}}^2$ for some $r_P > 0$. If $P_{\max}$ bounds that power, then

$$\left| \sum_i a_i b_i \right| \leq P_{\max} \left\| \mathcal{R}_P^{-1/2} \mathcal{K}_a \mathcal{R}_P^{-1/2} \right\|. \quad (7.6)$$

This follows by substituting $\mathcal{R}_P^{1/2} u$ in the quadratic form. Passivity alone does not give coercivity: a nearly lossless resonant source can have very different stored energy, amplitude and net consumed power. The source-effort bound is therefore not silently relabeled a power bound.

7.3 Why arbitrary source functions do not imply arbitrary loads

Let $\mathcal{Y} = L_0^2(\Gamma)$ be the real space of square-integrable zero-mean loads, equipped with the dimensionless norm $\|f\|_{\mathcal{Y}}^2 = \int_{\Gamma} |f|^2 dA / (A[\Gamma] \Pi_0^2)$, where $\Pi_0 > 0$ is a chosen stress scale. Define the continuous quadratic map $\mathcal{Q} : \mathcal{X} \rightarrow \mathcal{Y}$ by taking radiation traction and applying the zero-mean projection $\mathcal{P}_0$ of [equation \(2.12\)](#). The required load in this space is $b = \mathcal{P}_0(\sigma\kappa + \Delta\rho\Phi)$.

Proposition 7.1 (A conditional obstruction under bounded source effort). *Suppose $\mathcal{U}$ is Hilbert, $\mathcal{S}_{\Gamma} : \mathcal{U} \rightarrow \mathcal{X}$ is compact and linear, and $\mathcal{Q}$ is continuous. The set*

$$\mathfrak{R}_{C_u} = \{ \mathcal{Q}(\mathcal{S}_{\Gamma} u) : \|u\|_{\mathcal{U}} \leq C_u \} \quad (7.7)$$

is compact in $\mathcal{Y}$. If $\mathcal{Y}$ is infinite dimensional, this set has empty interior in $\mathcal{Y}$.

Proof. Any sequence in the closed source ball has a weakly convergent subsequence with limit in that ball. Compactness of $\mathcal{S}_\Gamma$ makes the corresponding traces converge strongly. Continuity of $\mathcal{Q}$ then makes the loads converge to the load of the limiting source. Thus the image is sequentially compact, hence compact in the normed load space. A compact subset of an infinite-dimensional normed space cannot contain an open ball: its closure would contain a scaled closed unit ball, which is not compact. $\square$

The hypothesis expresses spatial smoothing between regular sources and the interface. Positive source-interface separation, regular geometry and a frequency away from unresolved poles are settings in which such smoothing can be investigated; cylinder geometry alone does not prove compactness. Sources acting directly on the observed interface, singular distributions, changing source supports or an unbounded frequency range are not automatically covered. Nor does the result rule out a finite-dimensional Cartesian target family. It rules out claiming that bounded distributed actuation covers an open set of unrestricted load fields merely because the source has infinitely many degrees of freedom. This is a deduction for the stated operator, not a published universal no-go theorem about acoustic lenses.

7.4 A well-defined regularized fixed-target problem

Let $\gamma_u > 0$ be a dimensionless regularization weight. A reusable fixed-geometry inverse is

$$\min_{u \in \mathcal{U}_{\text{ad}}} j_\Gamma(u), \quad j_\Gamma(u) = \frac{1}{2} \|\mathcal{Q}(\mathcal{S}_\Gamma u) - b\|_{\mathcal{Y}}^2 + \frac{\gamma_u}{2} \|u\|_{\mathcal{U}}^2. \quad (7.8)$$

It minimizes load error and declared source effort. The positive penalty selects controlled effort; it may deliberately leave nonzero mismatch even when an exact match exists. A tolerance-constrained minimum-effort problem is another choice. Neither is generally convex because the load is quadratic in coherent source amplitude.

Proposition 7.2 (Conditional existence of a regularized source). *Under the compactness assumptions above, if $\mathcal{U}_{\text{ad}}$ is nonempty and weakly closed, equation (7.8) has a minimizer.*

Proof. A minimizing sequence has bounded norm because $\gamma_u > 0$ and the mismatch is nonnegative. Select a weakly convergent subsequence in $\mathcal{U}$. Its limit is admissible by weak closedness. Compact wave transfer and continuous quadratic loading give strong convergence of the load, while the squared source norm is weakly lower semicontinuous. The limit therefore attains the infimum. $\square$

This proof guarantees a minimizer of the declared reduced problem, not zero error, uniqueness, an algorithm finding the global minimum, a stable fluid state, or existence for the full moving-interface system. Nonconvex hardware constraints need their own weak-closedness check. Regularization and support restrictions are also central to linear Helmholtz source problems [11, 10]; the nonlinear load problem above has its own explicit assumptions and proof.

7.5 A second-order entry that needs no neighboring target

A concrete analytic construction addresses the zero-source degeneracy. Assume zero actuation is admissible. Choose M generalized forces and positive scales $F_{i,0}$ in newtons. For a positive dimensionless penalty γ_u , consider the projected cost

$$j_M(u) = \frac{1}{2} \sum_{i=1}^M \left(\frac{F_i(u) - b_i}{F_{i,0}} \right)^2 + \frac{\gamma_u}{2} \|u\|_{\mathcal{U}}^2. \quad (7.9)$$

Define the dimensionless self-adjoint operator

$$\mathcal{H}_0 = \gamma_u \mathcal{I}_{\mathcal{U}} - 2 \sum_i \frac{b_i}{F_{i,0}^2} \mathcal{K}_i, \quad (7.10)$$

where $\mathcal{I}_{\mathcal{U}}$ is the identity on the source space. Expanding the squares gives the exact identity

$$j_M(u) - j_M(0) = \frac{1}{2} \langle u, \mathcal{H}_0 u \rangle_{\mathcal{U}} + \frac{1}{2} \sum_i \left(\frac{F_i(u)}{F_{i,0}} \right)^2. \quad (7.11)$$

Thus $\mathcal{H}_0 \succeq 0$ makes zero a global minimizer of this particular penalized objective, even though it need not meet the target tolerance. This signals a limitation of the objective or available wave response, not proof that the target is physically impossible. If an admissible direction has negative quadratic form, a sufficiently small nonzero source in that direction strictly improves the cost. The real first derivative at zero cannot detect it.

There is also an exact amplitude optimization along any direction. Let $d \in \mathcal{U}$ have unit norm, write $u = \sqrt{x} d$ with $x \geq 0$, and define $f_i = F_i(d)/F_{i,0}$, $\hat{b}_i = b_i/F_{i,0}$, $A_d = \sum_i \hat{b}_i f_i - \gamma_u/2$, and $B_d = \sum_i f_i^2$. Then

$$j_M(\sqrt{x} d) = j_M(0) - A_d x + \frac{1}{2} B_d x^2, \quad x_* = \min\{C_u^2, \max\{0, A_d/B_d\}\} \quad (B_d > 0), \quad (7.12)$$

for the sole bound $\|u\|_{\mathcal{U}} \leq C_u$. If $B_d = 0$, the positive regularization makes $x_* = 0$. Additional physical constraints change the admissible interval or can invalidate the direction. A negative-curvature direction of $\mathcal{H}_0$ and this amplitude rule give a principled entry to a source solve for the target at hand. Optimizing the direction, enforcing the full unsampled load, and establishing fluid stability remain necessary; this construction is not a global inverse solution or a family-coverage proof.

7.6 What changes when the surface is also unknown

Let $\mathbf{y}$ collect the surface, hydrodynamic state, acoustic state and, where needed, temperature and actuator variables. Let $\mathcal{R}(\mathbf{y}, u) = 0$ collect the stationary governing equations, volume conditions, boundary conditions and chosen pressure gauges. Let $\Gamma_{\text{tar}} \subseteq \Gamma_*$ denote the target patch, of area $A_{\text{tar}} > 0$. For a nearby surface let η_s be its signed normal departure on this fixed target patch, and let $h_0 > 0$ be a geometric tolerance. Define the dimensionless shape error

$$J_s = \frac{1}{A_{\text{tar}} h_0^2} \int_{\Gamma_{\text{tar}}} \eta_s^2 dA. \quad (7.13)$$

Larger deformations need a declared surface-distance and correspondence convention. Let $J_{\text{obs}} \geq 0$ denote any additional dimensionless application-specific observation error, and choose nonnegative weights $\beta_s, \beta_o, \beta_u$. A coupled design can minimize

$$J_{\text{stat}} = \beta_s J_s + \beta_o J_{\text{obs}} + \beta_u \|u\|_{\mathcal{U}}^2 \quad \text{subject to } \mathcal{R}(\mathbf{y}, u) = 0. \quad (7.14)$$

The admissible set also imposes the declared source and stability limits. For an exact target impose $\Gamma = \Gamma_*$ directly; a soft penalty alone does not enforce exact shape. Optical observations are a later specialization. The full problem is nonlinear because both required curvature and acoustic transfer depend on geometry; the fixed-target relaxation does not make it globally convex.

Let δ denote a first variation of geometry, material state or drive. Differentiating [equation \(7.1\)](#) gives

$$\mathcal{A}_{\Gamma} \delta \mathbf{z}_a = (\delta \mathcal{B}_{\Gamma}) u + \mathcal{B}_{\Gamma} \delta u - (\delta \mathcal{A}_{\Gamma}) \mathbf{z}_a. \quad (7.15)$$

Moving normals, surface measure, interface conditions and material coefficients are part of this derivative. Freezing the acoustic field while differentiating only capillary curvature omits the central wave–shape feedback.

If the linearized constrained stationary operator $\mathcal{R}_y$ is invertible on the gauge-fixed state space, implicit differentiation gives

$$\delta y = -\mathcal{R}_y^{-1} \mathcal{R}_u \delta u. \quad (7.16)$$

At a fold or singular branch that inverse fails, and a local sensitivity cannot be interpreted as a global design map. Resonance can also make it very poorly conditioned.

Finally, let $\mathbf{a}$ be the actually commanded electrical or mechanical field and let $u = \mathcal{H}(\Gamma, \omega)\mathbf{a}$ be its loaded source response. Holding $\mathbf{a}$ fixed generally does not hold u fixed. The shape derivative includes $\delta\mathcal{H}$. Prescribed boundary velocity is experimentally relevant only when it is controlled or when the transducer dynamics needed to realize it are included. The same distinction applies to source gradients.

7.7 Finite arrays approximate a source space

Let $\mathcal{U}_N$ be a nested sequence of finite-dimensional source spaces. An array or basis represents $u = E_N \mathbf{w}$, where E_N maps its N complex coefficients to physical source fields. This is a restriction of [equation \(7.8\)](#), not its definition. Assume that every $u \in \mathcal{U}_{\text{ad}}$ admits strongly convergent admissible approximants $u_N \in \mathcal{U}_N \cap \mathcal{U}_{\text{ad}}$.

Proposition 7.3 (Conditional approximation of the inverse value). *Under the existence hypotheses and the admissible-approximation assumption, the minima of j_Γ over $\mathcal{U}_N \cap \mathcal{U}_{\text{ad}}$ converge to the minimum over $\mathcal{U}_{\text{ad}}$. Every weak accumulation point of corresponding minimizers is a minimizer of the continuous problem.*

Proof. The restricted infima cannot be smaller than the continuous infimum. Strong admissible approximants to a continuous minimizer have convergent objective values, giving the reverse limiting bound. Their bounded values bound the norms of the restricted minimizers. Apply the weak-compactness and lower semicontinuity argument of the preceding proof to their subsequences. $\square$

Density of an unconstrained basis alone is insufficient; the approximants must respect the physical bounds. Arbitrary unrelated arrays need not form such a sequence. This statement also assumes the exact same wave model; it does not replace a separate convergence analysis of acoustic or fluid meshes.

8 Finite-array restrictions and coherent-load relaxation

8.1 A linear wave response produces a quadratic load

We now restrict the distributed problem to a finite boundary-velocity array. Fix a geometry, material state and carrier frequency for which the linear wave problem is well posed, and set $\mathbf{F} = \mathbf{q} = 0$. Let N be the number of independent complex mechanical actuator channels. In one useful boundary representation, let $b_j(\mathbf{x})$ be the dimensionless normal-velocity pattern of channel j on Γ_a . The imposed peak velocity is $\sum_{j=1}^N b_j w_j$. The entries w_j of $\mathbf{w} \in \mathbb{C}^N$ have units of velocity and describe physical motion, not uncalibrated electrical voltages. The restriction map of the preceding section is $E_N \mathbf{w} = (\sum_j b_j w_j, 0, 0)$.

Let $\mathcal{A}_\Gamma(\omega)$ be the linear wave operator, including all interface and exterior conditions, and $\mathcal{B}_\Gamma^{(N)}(\omega)$ the restricted source operator $\mathcal{B}_\Gamma E_N$. Its acoustic state $\mathbf{z}_a$ satisfies

$$\mathcal{A}_\Gamma(\omega)\mathbf{z}_a = \mathcal{B}_\Gamma^{(N)}(\omega)\mathbf{w}. \quad (8.1)$$

For each phase define a row response $\mathbf{h}_{p\ell}(\mathbf{x})$ and a three-row velocity response $\mathbf{H}_{v\ell}(\mathbf{x})$ by $P_\ell = \mathbf{h}_{p\ell}\mathbf{w}$ and $\mathbf{V}_\ell = \mathbf{H}_{v\ell}\mathbf{w}$. The former has units Pa s m^{-1} and the latter is dimensionless. Substitution in equations (4.3) and (4.4) gives Hermitian matrices

$$\mathbf{K}_\ell(\mathbf{x}) = \frac{\mathbf{h}_{p\ell}^\dagger \mathbf{h}_{p\ell}}{4\rho_\ell c_\ell^2} - \frac{\rho_\ell}{4} \mathbf{H}_{v\ell}^\dagger \mathbf{H}_{v\ell} + \frac{\rho_\ell}{2} \mathbf{H}_{v\ell}^\dagger \mathbf{n} \mathbf{n}^T \mathbf{H}_{v\ell}, \quad (8.2)$$

$$\mathbf{K}(\mathbf{x}; \Gamma, \omega) = \mathbf{K}_- - \mathbf{K}_+, \quad \boxed{\Pi(\mathbf{x}) = \mathbf{w}^\dagger \mathbf{K}(\mathbf{x}; \Gamma, \omega) \mathbf{w}}. \quad (8.3)$$

A Hermitian matrix equals its conjugate transpose, so the load is real. The matrix $\mathbf{K}$ has units kg m^{-3} and need not be positive semidefinite for two interacting fluids. In the particular limit equation (4.6), it is positive semidefinite at each point. Tangential traction components also have Hermitian quadratic forms.

For a quiescent target, choose M real, dimensionless virtual-displacement functions ϕ_i compatible with the support and satisfying $\int_\Gamma \phi_i \, dA = 0$. Define generalized forces F_i , required forces b_i , and integrated matrices $\mathbf{K}_i$ by

$$\mathbf{K}_i = \int_\Gamma \phi_i \mathbf{K} \, dA, \quad F_i = \mathbf{w}^\dagger \mathbf{K}_i \mathbf{w}, \quad b_i = \int_\Gamma \phi_i (\sigma \kappa + \Delta \rho \Phi) \, dA. \quad (8.4)$$

Here $\mathbf{K}_i$ has units kg m^{-1} , while F_i, b_i are forces. The target equations are $F_i = b_i$ for all admitted variations. A finite collection is a necessary projection of the continuum balance; its unsampled complement and tangential balance still need verification. This construction is reusable for different targets. The wave response must, however, be updated when the target changes the scattering geometry.

8.2 Feasibility, coherence and a useful relaxation

Define the Hermitian drive matrix $\mathbf{W} = \mathbf{w} \mathbf{w}^\dagger$, whose entries have units $\text{m}^2 \text{s}^{-2}$. A matrix $\mathbf{W} \succeq 0$ is positive semidefinite if $\mathbf{q}^\dagger \mathbf{W} \mathbf{q} \geq 0$ for every complex vector $\mathbf{q}$. A nonzero deterministic coherent drive has rank $\mathbf{W} = 1$. Writing tr for trace, the exact fixed-target problem includes

$$\text{tr}(\mathbf{K}_i \mathbf{W}) = b_i, \quad \mathbf{W} \succeq 0, \quad \text{rank } \mathbf{W} \leq 1, \quad W_{jj} \leq w_{j,\max}^2. \quad (8.5)$$

The positive bounds $w_{j,\max}$ specify peak channel velocities. The rank-zero case is the undriven state. Let $F_{i,\text{ref}} > 0$ be meaningful force-error scales, let $P_{\text{ref}} > 0$ be a power scale, and let $\beta_P \geq 0$ weight power. If the passive, fixed-geometry actuator model gives average input power $P_{\text{in}} = \mathbf{w}^\dagger \mathbf{R}_P \mathbf{w}$, the Hermitian matrix $\mathbf{R}_P \succeq 0$ has units kg s^{-1} and includes the phasor averaging factor. A practical mismatch objective is

$$\min_{\mathbf{W}} \frac{1}{2} \sum_{i=1}^M \left[\frac{\text{tr}(\mathbf{K}_i \mathbf{W}) - b_i}{F_{i,\text{ref}}} \right]^2 + \beta_P \frac{\text{tr}(\mathbf{R}_P \mathbf{W})}{P_{\text{ref}}}, \quad (8.6)$$

subject to the positivity, rank and hardware constraints. For example, a pointwise peak pressure bound is linear in $\mathbf{W}$: $\text{tr}(\mathbf{h}_p^\dagger \mathbf{h}_p \mathbf{W}) \leq P_{\text{max}}^2$, where P_{max} is a declared pressure limit. Such bounds must hold throughout the specified domain, not merely at convenient sample points.

Dropping the rank constraint gives a convex semidefinite relaxation when all other constraints are convex at this fixed geometry. This is a standard quadratic-optimization construction [13].

Its objective value is a lower bound for the coherent problem, not proof that a physical coherent solution attains it. If an optimum has rank one, its factor supplies the drive up to a global phase. Explicitly, let $\lambda_1 > 0$ be the single nonzero eigenvalue and $\mathbf{q}_1$ a unit eigenvector of a rank-one optimum. Let A_j and φ_j denote channel j 's peak velocity and phase, and $v_{a,j}(t)$ its real velocity coefficient. They follow as

$$\mathbf{w} = \sqrt{\lambda_1} \mathbf{q}_1, \quad A_j = |w_j|, \quad \varphi_j = \arg w_j, \quad v_{a,j}(t) = A_j \cos(\omega t - \varphi_j). \quad (8.7)$$

The phase of a zero amplitude is immaterial. A higher-rank optimum requires additional work.

In particular, writing $\mathbf{W}$ as a sum of rank-one factors does not create independent same-frequency coherent fields: those fields add before their intensity is squared. A higher-rank $\mathbf{W}$ can represent a physical covariance $\mathbb{E}[\mathbf{w}\mathbf{w}^\dagger]$ only under an implemented incoherence or time-multiplexing scheme with appropriate averaging, bandwidth and peak limits. Covariance bounds constrain second moments; peak limits must additionally hold for each physically realized waveform. Switching among patterns must respect the acoustic and mechanical memory in [section 4](#). Different carrier frequencies generally have different $\mathbf{K}_i(\omega)$ and cannot be substituted into a fixed-frequency factorization without changing the problem.

Proposition 8.1 (A necessary load-feasibility certificate). *Let α_i be real coefficients and suppose $\sum_i \alpha_i \mathbf{K}_i \succeq 0$. Every exact feasible load vector in [equation \(8.5\)](#) must satisfy $\sum_i \alpha_i b_i \geq 0$.*

Proof. For $\mathbf{W} \succeq 0$, $\sum_i \alpha_i b_i = \text{tr}[(\sum_i \alpha_i \mathbf{K}_i) \mathbf{W}] \geq 0$. $\square$

A violated inequality rules out even the relaxed load under that wave model. The condition is only necessary; actuator bounds, coherence, optical compatibility and stability can still rule out a load that passes it. Its purpose is to expose unattainability before treating a failed optimizer as an algorithmic problem.

9 Continuous source synthesis and placement design

Let $g(\mathbf{y})$ be a complex normal source-velocity density on the admitted emitting surface Γ_s . For the complete chamber Green responses,

$$P_\ell(\mathbf{x}) = \int_{\Gamma_s} G_{p\ell}(\mathbf{x}, \mathbf{y}) g(\mathbf{y}) dA_y, \quad \mathbf{V}_\ell(\mathbf{x}) = \int_{\Gamma_s} \mathbf{G}_{v\ell}(\mathbf{x}, \mathbf{y}) g(\mathbf{y}) dA_y. \quad (9.1)$$

The responses include all transmission, wall, impedance and geometry effects; a free-space Green function is generally not interchangeable with them. Substitution in the radiation tensor gives the central continuous inverse

$$\mathcal{Q}_\Gamma[g](\mathbf{x}) = \int_{\Gamma_s} \int_{\Gamma_s} g(\mathbf{y})^* K_\Gamma(\mathbf{x}; \mathbf{y}, \mathbf{y}') g(\mathbf{y}') dA_y dA_{y'} = \Pi_{\text{req}}(\mathbf{x}). \quad (9.2)$$

There is one equation on each controlled face and one common source field. The Hermitian kernel is the pressure/velocity bilinear version of [equation \(8.3\)](#). Its sign, attainable range and bounded-effort limitations remain those of [section 7](#).

Finite emitters give $g_N = \sum_k a_k b_k(\mathbf{y}; \mathbf{r}_k, \ell_k)$, with support size ℓ_k and positions $\mathbf{r}_k = (R_s, \phi_k, z_k)$ on a cylinder. An axisymmetric restriction uses rings, but a ring-only control cannot in general correct non-axisymmetric disturbances. Position, support, frequency and commands can be jointly designed. If $e_i = \mathbf{a}^\dagger K_i \mathbf{a} - b_i$ and $J = \frac{1}{2} \sum_i (e_i / F_{i,0})^2$, then at fixed target

$$D_a J[\delta a] = 2 \text{Re} \sum_i \frac{e_i}{F_{i,0}^2} (K_i \mathbf{a})^\dagger \delta \mathbf{a}, \quad (9.3)$$

$$\partial_{r_{k,\alpha}} J = \sum_i \frac{e_i}{F_{i,0}^2} \mathbf{a}^\dagger (\partial_{r_{k,\alpha}} K_i) \mathbf{a}. \quad (9.4)$$

Constraint derivatives and source-effort penalties must be added. If changing placement also changes a reflecting or absorbing boundary, the derivative of the wave operator contributes; differentiating the source term alone misses it. Frequency derivatives likewise include material dispersion and impedances.

Point controls may be written formally as measures $\mu = \sum_k a_k \delta_{r_k}$ with a total-variation penalty, following measure formulations of Helmholtz inversion [10]. Their units, singular near fields and relation to finite emitter patches must be declared. Linear inverse-source recovery does not establish nonlinear traction synthesis. Source separation and control-region geometry materially affect the effort and conditioning of active field synthesis [11, 14].

Source refinement and field refinement are separate experiments. First freeze the actual source support, frequency and complex physical command function and refine the forward field and surface representation. Then compare nested source spaces under the same pointwise velocity and power bounds, each with its own independently checked forward solution. More source coefficients cannot compensate for an unresolved wave discretization. An optimized coarse field may select precisely the errors that vanish under refinement.

10 An explicit source adjoint with interface conditions

10.1 A traction objective on a fixed interface

We now derive spatial optimality equations, rather than leaving the inverse as an abstract residual. This section uses exactly the canonical transmission model of section 5. Its geometry is fixed during a source variation; the coupled shape derivative is a separate operation. The pressure trace P and normal-velocity trace v are continuous across the interface. Tangential velocity obeys $\mathbf{V}_{t\ell} = \nabla_\Gamma P / (i\omega\rho_\ell)$. Define three real material coefficients

$$c_P = \frac{1}{4} \left(\frac{1}{\rho_- c_-^2} - \frac{1}{\rho_+ c_+^2} \right), \quad c_v = \frac{\Delta\rho}{4}, \quad c_\tau = -\frac{1}{4\omega^2} \left(\frac{1}{\rho_-} - \frac{1}{\rho_+} \right). \quad (10.1)$$

Substitution into equations (4.3) and (4.4) gives

$$\boxed{\Pi = c_P |P|^2 + c_v |v|^2 + c_\tau |\nabla_\Gamma P|^2.} \quad (10.2)$$

All three terms have pressure units. The tangential radiation-traction jump vanishes in this inviscid transmission limit, since $\rho_\ell \mathbf{V}_{t\ell}$ and v are common on both sides. This is not a claim about thermoviscous boundary layers or variable surface tension.

Let Π_{tar} be a required load in pascals, $A_0 > 0$ a reference area, $\Pi_0 > 0$ a reference stress, and $\varpi \geq 0$ a smooth dimensionless weight supported away from the contact line. For a closed interface the last restriction is unnecessary. Choose positive source-penalty weights $\gamma_g, \gamma_F, \gamma_q$ for available components, using the scales of equation (5.2), and define

$$J_{\text{load}} = \frac{1}{2A_0\Pi_0^2} \int_\Gamma \varpi (\Pi - \Pi_{\text{tar}})^2 dA, \quad (10.3)$$

$$J_{\text{reg}} = \frac{\gamma_g}{2S_a g_0^2} \int_{\Gamma_a} |g|^2 dA + \frac{\gamma_F}{2V_s F_0^2} \int_{D_s} |\mathbf{F}|^2 dV + \frac{\gamma_q}{2V_s q_0^2} \int_{D_s} |q|^2 dV. \quad (10.4)$$

These costs are dimensionless. For a volume-constrained quiescent target use $\Pi_{\text{tar}} = \sigma\kappa + \Delta\rho\Phi - \lambda_V$ and optimize over λ_V as well. Its stationarity equation is the weighted removal of the otherwise arbitrary constant load. A weighted patch objective does not certify the unobserved part of the interface.

Define the real residual weight $r = \varpi(\Pi - \Pi_{\text{tar}})/(A_0\Pi_0^2)$ and the complex interface covectors

$$A_{\text{obs}} = 2c_P r P - 2\nabla_\Gamma \cdot (c_\tau r \nabla_\Gamma P), \quad B_{\text{obs}} = 2c_v r v. \quad (10.5)$$

Their units are $\text{Pa}^{-1} \text{m}^{-2}$ and s m^{-3} , respectively. At fixed target and geometry, differentiating the three squared moduli and integrating the tangential term by parts gives

$$DJ_{\text{load}}[\delta P, \delta v] = \text{Re} \int_{\Gamma} (\bar{A}_{\text{obs}} \delta P + \bar{B}_{\text{obs}} \delta v) dA. \quad (10.6)$$

For nonzero edge weights an additional term is $2 \text{Re} \int_{\mathcal{C}} c_r r(\mathbf{m} \cdot \nabla_{\Gamma} \bar{P}) \delta P ds$, where $\mathbf{m}$ is the outward unit co-normal tangent to Γ . It vanishes under the stated support restriction. Omitting it for a full-edge objective would require a justified edge condition, not a change of notation.

10.2 The adjoint has jumps even when pressure does not

Let Z_{ℓ} be a piecewise adjoint field; for the normalization above its units are $\text{s}^2 \text{m}^{-3}$, so it is not a physical pressure. Assume sufficient piecewise regularity for the following Green identity and traces. With ∂_n retaining the same orientation on both sides of Γ , solve

$$L_{\ell}^* Z_{\ell} = -\nabla \cdot (\alpha_{\ell} \nabla Z_{\ell}) - \bar{\beta}_{\ell} Z_{\ell} = 0 \quad \text{in } \Omega^{\ell}, \quad (10.7)$$

$$\alpha \partial_{n_a} Z + i\omega \bar{Y} Z = 0 \quad \text{on } \partial \mathcal{D}_c, \quad (10.8)$$

$$\boxed{[[Z]] = \frac{i}{\omega} B_{\text{obs}}, \quad [[\alpha \partial_n Z]] = -A_{\text{obs}}} \quad \text{on } \Gamma. \quad (10.9)$$

The bulk coefficient β_{ℓ} is real in the present canonical model; the conjugation records the formal-adjoint convention. A model with complex bulk coefficients also needs its consistently changed radiation and power expressions. The plus sign in the adjoint wall impedance follows from the chosen $e^{-i\omega t}$ convention.

To verify the interface signs, write $\delta S = -\nabla \cdot (\alpha \delta \mathbf{F}) - i\omega \delta q$, the volume right-hand side of the differentiated wave equation. Green's second identity, summed over the phases, gives

$$\int_{D_s} \delta S \bar{Z} dV = \int_{\Gamma} [-\delta P [[\alpha \partial_n \bar{Z}]] + i\omega \delta v [[\bar{Z}]]] dA - i\omega \int_{\Gamma_a} \delta g \bar{Z} dA. \quad (10.10)$$

The pressure variation is continuous and its common flux is $\alpha \partial_n \delta P = i\omega \delta v$. The two boxed jumps turn the interface integral into the complex expression whose real part is [equation \(10.6\)](#). Integrating the body-force source by parts, using its interior support, therefore yields

$$DJ_{\text{load}}[\delta u] = \text{Re} \left[i\omega \int_{\Gamma_a} \delta g \bar{Z} dA + \int_{D_s} \alpha \delta \mathbf{F} \cdot \nabla \bar{Z} dV - i\omega \int_{D_s} \delta q \bar{Z} dV \right]. \quad (10.11)$$

This derivation explains why an adjoint constrained to globally continuous Z would generally be wrong for an objective depending on normal velocity. An equivalent mixed or transposition formulation must encode these same interface functionals. Boundary terms are part of the derivative, as in continuous-adjoint methodology [\[15\]](#).

10.3 Spatial source gradients and necessary optimality

Define gradients using the unweighted real physical L^2 pairing:

$$D(J_{\text{load}} + J_{\text{reg}})[\delta u] = \text{Re} \left[\int_{\Gamma_a} \bar{G}_g \delta g dA + \int_{D_s} \bar{\mathbf{G}}_F \cdot \delta \mathbf{F} dV + \int_{D_s} \bar{G}_q \delta q dV \right]. \quad (10.12)$$

Combining [equations \(10.4\)](#) and [\(10.11\)](#) gives

$$\boxed{G_g = \frac{\gamma g g}{S_a g_0^2} - i\omega Z, \quad \mathbf{G}_F = \frac{\gamma_F \mathbf{F}}{V_s F_0^2} + \alpha \nabla Z, \quad G_q = \frac{\gamma q q}{V_s q_0^2} + i\omega Z.} \quad (10.13)$$

Each field is restricted to its own support. The gradient in the scaled Hilbert metric of [equation \(5.2\)](#) has the corresponding metric factors; the physical first variation is the same. If λ_V is free, also impose $\int_{\Gamma} r \, dA = 0$.

For unconstrained boundary actuation, the first-order condition is the implicit field equation

$$g = \frac{i\omega S_a g_0^2}{\gamma_g} Z|_{\Gamma_a}. \quad (10.14)$$

For the sole additional restriction $|g(\mathbf{x})| \leq g_{\max}(\mathbf{x})$, where $g_{\max} \geq 0$ is prescribed, this becomes pointwise projection of the right-hand side onto the complex disk of that radius. Phase-only control, source coupling or a spatial-derivative penalty does not have this simple projection rule. In a general convex admissible set the condition is that [equation \(10.12\)](#), with $\delta u = \tilde{u} - u$, is nonnegative for every $\tilde{u} \in \mathcal{U}_{\text{ad}}$.

These equations form a common state–adjoint–optimality system for every admissible target. They are not a closed-form answer: Z depends on the current pressure, load residual and geometry. The quadratic loading makes the objective nonconvex, so the system describes necessary local conditions. A vanishing source is always a stationary point for a pure quadratic-load tracking objective with quadratic regularization; a gradient-only iteration started exactly there can stop even when a useful nonzero source exists. That analytic degeneracy must be addressed by the eventual solution method, not concealed by a target-specific fit.

10.4 A limiting boundary check and extension to the full model

For a one-sided, fixed pressure-release surface, $P = 0$ and $\Pi = \rho_- |v|^2/4$. Let B_{obs} now be the corresponding velocity covector in [equation \(10.6\)](#). The exterior Green boundary term is $-i\omega \delta v \bar{Z}$, so the adjoint Dirichlet condition is

$$Z = -\frac{i}{\omega} B_{\text{obs}} \quad \text{on } \Gamma. \quad (10.15)$$

Setting $Z = 0$ there would lose the entire velocity-based objective derivative. This limit is a sign and boundary-condition check, not the assumed apparatus.

For the full coupled problem, the same integration-by-parts construction must include viscous and thermal fields, fast interface compliance, the mean flow, actuator loading and geometric transport. [Equation \(10.13\)](#) is complete for the explicitly stated frozen-geometry acoustic subproblem; it is not the full shape or thermoviscous adjoint. The general coupled and time-dependent adjoint equations are specified in [section 15](#).

11 Stability of a maintained interface

11.1 A stationary force balance is only the starting point

Let $(\mathbf{y}_*, \mathbf{a}_*)$ be a candidate maintained state and command. For an autonomous slow-state or acoustic-envelope description, define $\boldsymbol{\chi} = \mathbf{y} - \mathbf{y}_*$ in local coordinates satisfying the conservation constraints. Denote command perturbation by $\delta \mathbf{a} = \mathbf{a} - \mathbf{a}_*$, and the dynamic state and command derivative operators by $\mathbf{A}$ and $\mathbf{B}$, respectively. Linearization gives

$$\dot{\boldsymbol{\chi}} = \mathbf{A}\boldsymbol{\chi} + \mathbf{B}\delta \mathbf{a}. \quad (11.1)$$

The operators $\mathbf{A}, \mathbf{B}$ are derivatives of the dynamic equations, not derivatives of a surface-fitting routine. The state includes the necessary fluid, thermal, actuator and acoustic memory variables.

A steady phasor field represents a periodic physical wave; if an envelope reduction is unjustified, stability must instead be studied about that periodic state.

For a finite-dimensional autonomous reduction, negative real parts of all eigenvalues imply local linear asymptotic stability. For the continuum problem, point spectrum alone is insufficient: suitable evolution-operator bounds are needed. Define $\mathcal{T}(t)$ as the linear perturbation propagation operator, $M_s \geq 1$ as a dimensionless amplification bound, and $\alpha_s > 0$ as a decay rate. A stronger useful condition is

$$\|\mathcal{T}(t)\| \leq M_s e^{-\alpha_s t}, \quad t \geq 0. \quad (11.2)$$

The norm uses physically stated state scales. A large M_s permits substantial transient growth despite eventual decay. Acceptability additionally depends on the observation $\mathbf{C}_{\text{obs}}\boldsymbol{\chi}$, where $\mathbf{C}_{\text{obs}}$ is the linearized application-specific observation operator, and on the amplitude of admitted disturbances. Numerical convergence, shape accuracy and dynamical stability are three different claims.

11.2 Linearize the actual base flow in three dimensions

If the maintained surface carries a steady streaming flow $\mathbf{U}_\star$, then $\mathbf{U}_\star \cdot \mathbf{n} = 0$ on the surface does not imply $\mathbf{U}_\star = 0$ in the fluids. For constant-density mean momentum, the convective derivative contributes

$$\delta[(\mathbf{U} \cdot \nabla)\mathbf{U}] = (\mathbf{U}_\star \cdot \nabla)\delta\mathbf{U} + (\delta\mathbf{U} \cdot \nabla)\mathbf{U}_\star. \quad (11.3)$$

Both terms, incompressibility, perturbed viscous stress, moving-interface conditions and the derivatives of acoustic forcing belong in $\mathbf{A}$. Temperature-dependent coefficients and source loading contribute further terms when present. A spectrum obtained after setting this measured or computed base circulation to zero is the spectrum of a different problem.

The cylindrical container admits angular perturbations of every integer order m . If the complete base state and boundary laws are rotationally invariant, $\mathbf{A}$ commutes with rotations and can be decomposed into sectors proportional to $e^{im\theta}$. Stability requires control of all admissible sectors, including $m \neq 0$, not just the axisymmetric sector. An asymmetric base state couples these orders and needs the corresponding three-dimensional linearized operator. Conservation constraints, contact-line motion and legitimate symmetry modes must be handled before interpreting zero eigenvalues. A finite angular truncation also needs a bound or a convergence argument for its omitted modes.

11.3 Capillary stiffness and acoustic feedback

In the quiescent, constant- σ reduction, define the shape operator $\mathbf{B}_\Gamma = \nabla_\Gamma \mathbf{n}$ and its squared norm $|\mathbf{B}_\Gamma|^2$, equal to the sum of squared principal curvatures. Let $\Delta_\Gamma = \nabla_\Gamma \cdot \nabla_\Gamma$ denote the surface Laplacian. Under normal displacement η ,

$$\delta\kappa = -(\Delta_\Gamma + |\mathbf{B}_\Gamma|^2)\eta, \quad \delta\Phi = (\mathbf{n} \cdot \nabla\Phi)\eta. \quad (11.4)$$

At fixed actual commanded variable, denote the complete shape derivative of acoustic normal load by $D_\Gamma \Pi[\eta]$. The restoring-pressure operator, before projection onto volume-preserving variations, is

$$\boxed{\mathcal{K}_{\text{eff}}\eta = -\sigma(\Delta_\Gamma + |\mathbf{B}_\Gamma|^2)\eta + \Delta\rho(\mathbf{n} \cdot \nabla\Phi)\eta - D_\Gamma \Pi[\eta].} \quad (11.5)$$

The pressure-multiplier variation is removed on that constrained space. Boundary terms and contact-line perturbations must be included for the chosen wetting model. Thermal variations add derivatives of σ , density and other properties. This formula shows the relevant question: when the interface moves, does the resulting acoustic load counteract or reinforce the displacement? Matching the load at one geometry supplies no answer to that derivative.

If fluid inertia and acoustic memory are negligible, let $\mathcal{M}_\Gamma$ be the viscous mobility mapping outward traction to interface normal velocity, with its constraints and boundary conditions. Then the creeping-flow limit is

$$\dot{\eta} = -\mathcal{M}_\Gamma \mathcal{K}_{\text{eff}} \eta. \quad (11.6)$$

For passive Stokes flow the constrained mobility is positive in the reciprocal work pairing. A coercive self-adjoint restoring operator can then give an energy-based stability criterion. Driven, dissipative acoustic feedback need not be self-adjoint or derive from a scalar shape potential; positivity of a guessed “capillary plus acoustic energy” is not a general stability proof.

With memory retained, let $s \in \mathbb{C}$ be a Laplace frequency, $\mathcal{Z}_h(s)$ the hydrodynamic restoring/dynamic impedance, and $\mathcal{K}_a(s)$ the linearized acoustic load response to surface displacement. The homogeneous perturbation condition is

$$[\mathcal{Z}_h(s) - \mathcal{K}_a(s)]\eta = 0. \quad (11.7)$$

Both operators include their relevant boundary and state coupling. In a particular modal approximation, hydrodynamics may be represented by $\mathcal{Z}_h(s) = s^2 \mathbf{M} + s \mathbf{D} + \mathbf{K}_h$, where $\mathbf{M}, \mathbf{D}, \mathbf{K}_h$ are the modal mass, damping and passive stiffness matrices. Frequency-independent damping is an approximation, not a universal viscous free-surface law; capillary-wave dispersion depends on the viscous regime [16]. Eliminating acoustic state instantaneously replaces $\mathcal{K}_a(s)$ by its zero-frequency value and can miss an unstable oscillatory mode.

For a periodically maintained state of period T_p , let $\mathcal{M}_p$ denote the linear evolution over one period. In a finite-dimensional periodic reduction, its Floquet multipliers must lie strictly inside the unit circle after genuine symmetry modes are treated appropriately. A continuum statement also requires operator control. Pulse trains, modulation and beat forcing may therefore need a periodic analysis even when their mean load is correct.

11.4 Feedback must have the necessary authority and observations

Let the measured perturbation be $\mathbf{z}_{\text{obs}} = \mathbf{C}\boldsymbol{\chi} + \boldsymbol{\nu}$, where $\mathbf{C}$ is the sensor operator and $\boldsymbol{\nu}$ measurement noise. The existence of a desired equilibrium does not imply that its unstable modes can be controlled or observed. Local stabilization requires, at minimum, authority over the unstable modes and sufficient information to estimate those that need feedback. Delays, saturation and finite sensing bandwidth belong to the control model.

One precise necessary test uses a left unstable mode. Suppose an isolated mode ψ of the adjoint state operator satisfies $\mathbf{A}^* \psi = \bar{\lambda} \psi$, with $\text{Re } \lambda \geq 0$ and $\mathbf{B}^* \psi = 0$. In the usual complex pairing, its amplitude $a_\psi = \langle \psi, \boldsymbol{\chi} \rangle$ satisfies

$$\dot{a}_\psi = \lambda a_\psi + \langle \mathbf{B}^* \psi, \delta \mathbf{a} \rangle = \lambda a_\psi. \quad (11.8)$$

No linear feedback through the admitted source operator can make this mode decay exponentially. For an infinite-dimensional boundary-control problem, the adjoint pairing requires its appropriate trace and admissibility domain; the statement is for modes on which that pairing is defined. Similarly, an unstable state mode invisible to the measurements obstructs its asymptotic estimation by a detectable linear observer. These tests connect source support and sensing to stability, independently of a target-fitting algorithm.

For illustration only, consider real, scaled finite-dimensional coordinates and full-state feedback $\delta \mathbf{a} = -\mathbf{K}_c \boldsymbol{\chi}$. Here $\mathbf{K}_c$ is a feedback gain, distinct from the traction matrices $\mathbf{K}_i$. Let $\mathbf{P}_L \succ 0$ be a symmetric Lyapunov matrix. A sufficient linear decay condition is

$$(\mathbf{A} - \mathbf{B}\mathbf{K}_c)^T \mathbf{P}_L + \mathbf{P}_L (\mathbf{A} - \mathbf{B}\mathbf{K}_c) \preceq -2\alpha_s \mathbf{P}_L. \quad (11.9)$$

With $\mathbf{X} = \mathbf{P}_L^{-1}$ and $\mathbf{Y} = \mathbf{K}_c \mathbf{X}$, this becomes the linear matrix inequality

$$\mathbf{A}\mathbf{X} + \mathbf{X}\mathbf{A}^T - \mathbf{B}\mathbf{Y} - \mathbf{Y}^T\mathbf{B}^T + 2\alpha_s\mathbf{X} \preceq 0, \quad \mathbf{X} \succ 0. \quad (11.10)$$

This established construction [17] is convex for fixed $\mathbf{A}, \mathbf{B}, \alpha_s$ and the stated full-state model. Joint design of geometry, nonlinear acoustics and output feedback does not inherit that convexity. A controller satisfying it still needs a compatible estimator, actuator headroom and a nonlinear region of attraction containing the intended formation endpoint and disturbances.

11.5 What is needed to extend a linear certificate locally

A conditional estimate makes the limitation of a linear result concrete. In scaled state coordinates suppose the closed-loop nonlinear perturbation equation has the form

$$\dot{\mathbf{x}} = \mathbf{A}_c \mathbf{x} + \mathcal{N}(\mathbf{x}), \quad \|\mathcal{N}(\mathbf{x})\| \leq c_N \|\mathbf{x}\|^2 \quad \text{for } \|\mathbf{x}\| \leq r_0. \quad (11.11)$$

Here $\mathbf{A}_c = \mathbf{A} - \mathbf{B}\mathbf{K}_c$, $c_N \geq 0$ is a remainder bound, and $r_0 > 0$ is a neighborhood radius. Assume local well-posedness there and a bounded positive Lyapunov operator $\mathbf{P}_L$ for which equation (11.9) holds as a quadratic-form inequality on the generator domain, replacing matrix transpose by the Hilbert adjoint when using an operator state space. Let $p_-, p_+ > 0$ satisfy $p_- \|\mathbf{x}\|^2 \leq V_L \leq p_+ \|\mathbf{x}\|^2$, where $V_L = \langle \mathbf{x}, \mathbf{P}_L \mathbf{x} \rangle$. For classical solutions,

$$\dot{V}_L \leq -2\alpha_s V_L + 2p_+ c_N \|\mathbf{x}\|^3. \quad (11.12)$$

If $c_N > 0$, set $r_* = \min\{r_0, \alpha_s p_- / (2p_+ c_N)\}$; if $c_N = 0$, take $r_* = r_0$. For $\|\mathbf{x}\| \leq r_*$ the positive term is at most $\alpha_s V_L$, hence $\dot{V}_L \leq -\alpha_s V_L$. Initial data with $V_L(0) < p_- r_*^2$ remain in that neighborhood and decay, as long as the stated solution and control constraints remain valid.

This is a local conditional estimate, not a claim that the free-boundary equations satisfy the assumed quadratic remainder in an arbitrary norm. In a continuum problem, regularity and possible derivative loss require care in choosing that state space. Formation must end inside a verified admissible neighborhood, and source saturation or disturbances require their own bounds. A negative finite-matrix spectrum by itself supplies none of these nonlinear hypotheses.

12 A two-interface inertial formation diagnostic

The following is a deliberately restricted forward model, not a replacement for the viscous, streaming, thermal and three-dimensional problem above. Assume axisymmetric, incompressible, irrotational mean motion, impermeable slip walls, and small interface displacement relative to each layer depth. Evaluate the kinetic energy on the flat reference domains. Retain the nonlinear graph capillary energy and recompute harmonic radiation loads on the instantaneous graphs. No viscous damping or settling law is introduced.

Here is a direct derivation of the inertial coupling. For a radial mode $J_0(kr)$, with $J_1(kR) = 0$, the potential in each layer satisfies $\phi_{zz} - k^2 \phi = 0$. The zero mode is excluded by the sealed-volume constraints. If the bottom and top vertical velocities of a layer of depth h are u and v , respectively, its potential is

$$\phi(z) = \frac{v - u \cosh(kh)}{k \sinh(kh)} \cosh(kz) + \frac{u}{k} \sinh(kz), \quad 0 \leq z \leq h.$$

Integration of $\rho |\nabla \phi|^2 / 2$, or Green's identity, gives kinetic energy per radial modal norm

$$\frac{\rho}{2k} [\coth(kh)(u^2 + v^2) - 2uv \operatorname{csch}(kh)].$$

Consequently, the two-interface modal mass is

$$M_k = \frac{1}{k} \begin{pmatrix} \rho_0 c_0 + \rho_1 c_1 & -\rho_1 s_1 \\ -\rho_1 s_1 & \rho_1 c_1 + \rho_2 c_2 \end{pmatrix}.$$

Here $c_j = \coth(kh_j)$ and $s_j = \operatorname{csch}(kh_j)$. Projection of the pinned, volume-preserving spline velocities into these radial modes yields a symmetric positive mass matrix M . This projection must be converged independently of the acoustic mesh. An optional Ritz restriction retains the lowest eigenvectors of $K_0 e = \omega^2 M e$; it is a declared dynamical resolution, not a source optimization.

With spline coordinates q , or their Ritz restriction, the forward equation is

$$M\ddot{q} + \nabla E(q) = F_{\text{ac}}(q, a(t)g), \quad q(0) = \dot{q}(0) = 0.$$

The target does not occur on the right-hand side. A specified source envelope $a(t)$ is a physical command, not interpolation towards a target. With zero drive, the continuous reduced model conserves $\dot{q}^T M \dot{q}/2 + E(q)$. For nonzero drive its derivative is $\dot{q}^T F_{\text{ac}}$; no convergence to an equilibrium is implied. A midpoint mechanical update with a predicted midpoint acoustic solve is second order locally and must be checked at fixed commands by time-step refinement. Large deformation, omitted modes, unresolved wave fields and resonant sensitivity can invalidate this diagnostic. Neither its error curve nor its optical rendering establishes the ten-nanometre criterion of the full apparatus.

13 Determine the load before synthesizing emitters

An emitter-independent test isolates the mechanical question: if the required mean traction could be supplied exactly, would the proposed pair be an equilibrium, and what would make it form and remain there? This is an ideal distributed-load problem, not evidence that an acoustic field realizes it. The following deductions specialize the variational mechanics of [sections 2 and 3](#); they introduce no empirical settling law or additional material data.

13.1 Two full-domain loads with an explicit sign convention

Let D be the entire horizontal disk, including both non-optical annuli. Number the three layers from bottom to top by $0, 1, 2$, and write their two interfaces as $z = h_j(x, y)$, $j = 0, 1$. Use upward normals on *both* interfaces, not the outward normals of the middle lens. Define $b_j = (\rho_j - \rho_{j+1})g_{\text{grav}}$ and let f_j be upward normal mean traction, in Pa. Its virtual work is $\int_D f_j \delta h_j \, dx \, dy$: the graph area factor cancels the normal projection of vertical displacement. For sealed outer layers, each $\int_D h_j \, dx \, dy$ is fixed; the middle volume follows from their difference. For compatible full targets h_j^* ,

$$f_j^*(x, y) = -\sigma_j \nabla \cdot \frac{\nabla h_j^*}{\sqrt{1 + |\nabla h_j^*|^2}} + b_j h_j^* - \lambda_j, \quad (x, y) \in D. \quad (13.1)$$

The two constants are hydrostatic pressure-difference multipliers, not independently prescribed absolute pressures. Subtracting each disk average is one reporting gauge; absolute pressure safety remains a separate issue. Changing the reference height only changes that gauge. For outward lens tractions the front sign reverses. Neither load may be silently set to zero outside the optical pupil. Different admissible annuli give different loads, even with identical optical conjugates. No unique numerical pressure map exists until the complete geometry, fluids, fill and support are specified.

13.2 A conditional three-dimensional stability result for ideal loads

Hold $f_j^*(x, y)$ fixed as a function of horizontal position while the interfaces move. Define the loaded potential

$$\mathcal{U}[h] = \sum_{j=0}^1 \int_D \left[\sigma_j \sqrt{1 + |\nabla h_j|^2} + \frac{1}{2} b_j h_j^2 - f_j^* h_j \right] dx dy. \quad (13.2)$$

Suppose $\sigma_j > 0$, $b_j \geq 0$, the boundary heights are pinned, and admissible graphs satisfy the fixed volumes and positive layer gaps. The second variation in [equation \(3.4\)](#), summed over both interfaces, is strictly positive for any nonzero admissible perturbation. Indeed, its capillary integrand is bounded below by $\sigma_j |\nabla \eta_j|^2 / (1 + |\nabla h_j|^2)^{3/2}$. The pinned boundary excludes constant perturbations. The admissible graph set is convex, so h^* is its unique stationary point and strict global minimizer of $\mathcal{U}$. This statement allows arbitrary $\eta_j(x, y)$, including every angular order, not just axisymmetric disturbances. Uniform coercivity in a neighborhood additionally uses a bound on target slopes. It is not a result for overturning interfaces or a density-inverted stack.

For smooth incompressible Newtonian mean flow with passive stationary walls, constant surface tensions, no streaming or other bulk forcing, and exactly these imposed normal tractions, the mechanical balances give

$$\frac{d}{dt} \left[\sum_{\ell=0}^2 \frac{1}{2} \int_{\Omega_\ell(t)} \rho_\ell |\mathbf{U}_\ell|^2 dV + \mathcal{U}[h] - \mathcal{U}[h^*] \right] = - \sum_{\ell=0}^2 \int_{\Omega_\ell(t)} 2\mu_\ell \mathbf{D}(\mathbf{U}_\ell) : \mathbf{D}(\mathbf{U}_\ell) dV \leq 0. \quad (13.3)$$

There is no invented diagonal damping matrix in this identity. Pressure constraint work vanishes by volume conservation; the two interfaces remain coupled through the fluid velocity and pressure fields. With sufficient regularity and coercivity it supplies energy stability about the target. Asymptotic convergence additionally needs dissipation that excludes persistent motions, global admissible solutions and compactness or decay estimates. An energy norm does not alone certify maximum height or optical slope errors. With zero viscosity, the right side vanishes: oscillation is allowed and settling cannot be claimed. Thus an inviscid animation cannot answer the user's maintained-formation question even with perfect loading.

The current hypothetical densities 1200, 1100, 1000 kg m⁻³ give $b_0 = b_1 = 981 \text{ Pa m}^{-1}$ at $g_{\text{grav}} = 9.81 \text{ m s}^{-2}$, satisfying the sign hypothesis. These are not characterized laboratory liquids. The result does not transfer automatically to a lens surrounded by one common liquid: then the two density jumps have opposite signs unless densities match.

13.3 Formation requires a load history, not just its endpoint

Applying f^* from a flat initial state defines a genuine forward initial value problem. It does not prescribe intermediate shapes. For a time-varying load, the energy balance includes its time-dependent external work; no monotonic target-error curve follows from a ramp. A full prescribed trajectory would instead require solving its compatible velocity and pressure fields and recovering the interfacial traction from their stress jump, including inertia and viscosity. It cannot generally be obtained by evaluating only the static curvature formula along an interpolated shape.

In the specifically declared constant-mass reduction of [section 12](#), the analogous inverse dynamic force is $F_{\text{req}}(t) = M\ddot{q}_d + \nabla E(q_d)$; a viscous resistance may be added only after it is derived or independently validated. Endpoint balance removes the acceleration term but says nothing about transient overshoot. This identity defines a diagnostic load, not an acoustic realization or a forward-verified trajectory.

13.4 Verification gates before returning to source design

First export capillary, gravitational and gauge-fixed total loads on both full targets, with geometry, volume and pinning checks. Independently solve the nonlinear prescribed-load equilibrium from a non-target initial guess; refine both the load representation and the mechanical grid. Next solve viscous two-interface formation with independently specified material data and initial conditions, refine spatial and temporal resolution, and test three-dimensional disturbances. Report both maximum height errors, slopes, fixed-detector maximum spot radius and lost rays throughout physical time. Distinguish a conditional ideal-load pass from any acoustic pass.

Only then ask the acoustic inverse to realize the loads and the required response to perturbations. At fixed emitter commands the traction is $f[h, a]$, not the fixed $f^*(x, y)$ above; its shape derivative changes the restoring operator and invalidates the ideal-load Lyapunov proof unless controlled. Mean tangential forcing, bulk streaming, wall layers and thermal feedback also require their own balances. The required acoustic specification is therefore a full-domain equilibrium load *plus* an admissible dynamic response, not a single pressure-amplitude image on the pupils.

14 Variational load inversion with coupled viscous fluids

14.1 The load is a multiplier of a physical evolution constraint

Retain three incompressible Newtonian phases with positive viscosities, piecewise constant densities, constant interfacial tensions, and pinned contact lines at stationary no-slip walls. The interfaces may depend on both horizontal coordinates. This subsection first excludes acoustic bulk forcing, streaming and thermal evolution to define a mechanically realizable ideal-load test. Those effects are restored as separate terms below, not claimed absent in the eventual apparatus. Let $G_h \mathbf{U}$ be the graph kinematic map: $(G_h \mathbf{U})_j = U_z - \nabla h_j \cdot \mathbf{U}_{xy}$ on interface j . Use $\langle f, v \rangle_D = \sum_j \int_D f_j v_j \, dx \, dy$. The velocity is continuous across both interfaces, divergence-free in each phase, and vanishes at the walls. Define the viscous bilinear form

$$a_h(\mathbf{U}, \mathbf{W}) = \sum_{\ell=0}^2 \int_{\Omega_\ell(h)} 2\mu_\ell \mathbf{D}(\mathbf{U}) : \mathbf{D}(\mathbf{W}) \, dV.$$

It has units of power when both arguments are velocities. The dissipation potential is half this quadratic form. A Rayleighian construction of this kind underlies viscous variational models [6]; here the contact line is fixed and no moving-contact-line law is introduced.

For a supplied graph velocity v_d compatible with all sealed volumes, consider the saddle functional

$$\mathcal{R} = \frac{1}{2} a_h(\mathbf{U}, \mathbf{U}) + \langle E'_h, G_h \mathbf{U} \rangle_D - \sum_\ell \int_{\Omega_\ell} p_\ell \nabla \cdot \mathbf{U} \, dV - \langle f, G_h \mathbf{U} - v_d \rangle_D. \quad (14.1)$$

Here E'_h is the full capillary–gravity variation. Variation in p and f enforces incompressibility and the desired interface speed. Variation in $\mathbf{U}$ gives the three-fluid Stokes weak equation. Thus f is the total required normal load, not a guessed pressure amplitude. Tangential traction continuity is a natural condition of this scalar normal-load problem. If a desired evolution requires incompatible tangential or bulk forces, normal pressure alone does not realize it.

On the admissible volume-preserving graph-velocity space define

$$\mathcal{D}_h(v) = \min_{\nabla \cdot \mathbf{U}=0, G_h \mathbf{U}=v} \frac{1}{2} a_h(\mathbf{U}, \mathbf{U}) = \frac{1}{2} \langle v, \mathcal{Z}_h v \rangle_D. \quad (14.2)$$

Existence and the operator domain require suitable trace regularity and compatible boundary data. Reciprocity and nonnegative dissipation imply a symmetric positive resistance $\mathcal{Z}_h$ on that

domain. Its off-diagonal blocks couple the two interfaces through the middle fluid. Where its inverse exists, $\mathcal{M}_h = \mathcal{Z}_h^{-1}$ is the Stokes mobility. The load and forward equations are respectively

$$\boxed{f_{\text{req}} = E'_h + \mathcal{Z}_h v_d - \lambda, \quad \dot{h} = \mathcal{M}_h(f - E'_h).} \quad (14.3)$$

Pressure constants are understood modulo the volume constraints. Neither a diagonal relaxation constant nor an independently moving front and back is consistent with this construction. At $v_d = 0$ the viscous term vanishes, recovering exactly the earlier static load. Adding viscosity does not change a quiescent equilibrium pressure requirement; it changes the approach to it.

14.2 What changes when inertia and additional forcing are retained

With material acceleration retained, the inverse weak equation for a desired admissible fluid motion is

$$\begin{aligned} \langle f - E'_h, G_h \mathbf{W} \rangle_D &= \sum_{\ell} \int_{\Omega_{\ell}} \rho_{\ell} (\partial_t \mathbf{U} + \mathbf{U} \cdot \nabla \mathbf{U}) \cdot \mathbf{W} \, dV + a_h(\mathbf{U}, \mathbf{W}) \\ &\quad - \sum_{\ell} \int_{\Omega_{\ell}} p_{\ell} \nabla \cdot \mathbf{W} \, dV - \mathcal{W}_{\text{additional}}(\mathbf{W}). \end{aligned} \quad (14.4)$$

The last term is the power pairing of any additional mean bulk and tangential loads. A scalar f exists only if the right-hand side factors through G_h : it must vanish for every admissible test velocity with $G_h \mathbf{W} = 0$ after the bulk equations and pressure are solved. This is a necessary compatibility condition, not automatic controllability of a prescribed shape history. Hydrodynamic memory means that shape and its instantaneous velocity alone need not determine the required load. If the endpoint is a streaming state, convective and viscous stresses persist there. With heating, density and surface tension change and Marangoni stresses appear; the constant-property static map is no longer the complete required acoustic load.

14.3 A forward test that does not prescribe its answer

For the ideal-load diagnostic choose the fixed full-domain static map $f^* = E'_{h^*} - \lambda$ and start the forward problem at flat compatible interfaces. The target appears only in this fixed load, not in a time-varying shape prescription. In the Stokes limit its loaded potential obeys

$$\frac{d\mathcal{U}}{dt} = -\langle E'_h - f^*, \mathcal{M}_h(E'_h - f^*) \rangle_D \leq 0.$$

For a finite-dimensional volume-constrained discretization with bounded, positive mobility and a strongly convex loaded potential on an invariant admissible set, convergence to its unique minimizer follows. A continuum or large-deformation assertion needs the corresponding regularity, gap and coercivity bounds; a computed time series is not their proof.

The first numerical implementation uses axisymmetric Taylor–Hood velocity and pressure elements on a moving, interface-fitted three-layer mesh. Velocity is continuous, pressure is continuous within each phase but allowed to jump at the interfaces. Only one global pressure gauge is removed. Both interface volume fluxes follow from the phasewise divergence constraints. The full axisymmetric strain includes hoop strain. The graph kinematics are projected into pinned volume-preserving spline coordinates; the transpose of the same map transfers generalized loads into the fluid. This reciprocal pairing preserves the discrete work identity. The mesh represents curved interfaces by line segments and must be refined separately from the spline space.

This first implementation is creeping flow: inertia, non-axisymmetric flow, streaming and heat transport are not implemented by it. Viscosities are explicit scenario inputs, not measured

values inferred from desired settling. Report velocity-based Reynolds estimates and a viscous diffusion-time check before using it for a physical formation claim. In particular an abrupt load switch in the Stokes model has no resolved inertial startup. Time refinement, fluid-grid refinement, independent analytic flow limits and energy checks are required. A pressure-history or array design cannot be declared sufficient merely because this restricted diagnostic reaches its endpoint.

15 Formation as constrained optimal control

15.1 State, control and a dimensionally defined objective

Let $\boldsymbol{\vartheta}$ collect fixed but possibly uncertain material and apparatus parameters. The symbol is distinct from temperature Θ and contact angle θ . Let $\mathcal{D}$ denote the full dynamic residual, including bulk equations, moving-interface laws, boundary conditions and algebraic constraints. For a prescribed initial physical state $\mathbf{y}_0$, write

$$\mathcal{D}(\mathbf{y}, \dot{\mathbf{y}}, \mathbf{a}; \boldsymbol{\vartheta}) = 0, \quad \mathbf{y}(0) = \mathbf{y}_0, \quad 0 \leq t \leq T_f. \quad (15.1)$$

This notation can represent the full compressible problem or a declared averaged reduction. In the former the command is the actual waveform; in the latter it may be its slowly varying envelope. They are not interchangeable for an arbitrarily short pulse.

Use the geometric error J_s defined in [equation \(7.13\)](#) on a general target patch. Let J_{obs} be the declared dimensionless observation objective, such as error in measured shape or flow. Cartesian optical objectives will specialize it in [section 16](#); they are not needed to define the shaping problem. For large departures use a declared geometric distance and correspondence, rather than implicitly registering a target to the evolving state.

Let $\mathbf{S}_a$ be an invertible scaling operator on the commanded fields (for a finite array, a diagonal matrix of positive channel scales) and $T_{\text{ref}} > 0$ a rate scale in seconds. Let $\beta_o, \beta_s, \beta_a, \beta_{\dot{a}} \geq 0$ be dimensionless weights and $\Phi_T(\mathbf{y}(T_f))$ a dimensionless terminal objective. A concrete family of problems is

$$\min_{\mathbf{a}, \mathbf{y}} \quad J = \Phi_T(\mathbf{y}(T_f)) + \int_0^{T_f} \ell(\mathbf{y}, \mathbf{a}, \dot{\mathbf{a}}, t) dt, \quad (15.2)$$

$$\ell = \frac{1}{T_f} \left[\beta_o J_{\text{obs}} + \beta_s J_s + \beta_a \left\| \mathbf{S}_a^{-1} \mathbf{a} \right\|^2 + \beta_{\dot{a}} \left\| T_{\text{ref}} \mathbf{S}_a^{-1} \dot{\mathbf{a}} \right\|^2 \right], \quad (15.3)$$

subject to [equation \(15.1\)](#) and the physical constraints below. The norms are the declared Hilbert norms of the scaled spatial command fields, integrated over their source supports; they reduce to Euclidean norms only after discretization. The penalties describe command effort and smoothness; they are not automatically acoustic power. A calibrated input-energy term can replace or supplement them. For a velocity-controlled fluid boundary with domain-outward normal $\mathbf{n}_a$ and wall velocity $\mathbf{v}_a$, instantaneous mechanical power entering that fluid is $\int_{\Gamma_a} (\mathbf{T} \mathbf{n}_a) \cdot \mathbf{v}_a dA$. Electrical power requires the electromechanical source model. Reversible energy returned to a source and gross consumed power are different quantities.

Terminal conditions depend on the purpose. For capture after a pulse, choose Φ_T to penalize optical error, residual fluid kinetic energy and remaining acoustic energy, each divided by a declared positive energy or optical scale. Require the command to end at zero and apply the retention condition in [equation \(2.13\)](#). For driven operation, target the complete holding state instead; its required nonzero acoustic field is not an unwanted residual. For curing, add conversion, heat release, shrinkage, evolving rheology and the final optical index before claiming a retained solid lens.

15.2 Constraints carry physical meaning

The admissible set must specify the following quantities, according to the chosen physical model:

1. conserved masses or supplied volume constraints, support and wetting conditions, clearance and an admissible non-self-intersecting interface;
2. command amplitude, bandwidth, rate, total energy, source loading and the range in which its acoustic response remains valid;
3. acceptable temperature and flow, and any pressure, cavitation or damage limits supported by material and apparatus data;
4. optical path and direction tolerances at the requested conjugates, transmission without total internal reflection, and the intended aperture, wavelength and exposure interval;
5. stability, disturbance rejection and actuator headroom during the specified holding interval.

There is no universal cavitation threshold or acceptable intensity independent of liquid, nuclei, frequency and exposure. Such a constraint must come from a specified model or measurement. Similarly, optical quality after curing cannot be certified with only liquid-state refractive indices.

15.3 Adjoint equations give a general derivative method

The inverse method need not be a new heuristic for each target. A common constrained state equation and a common objective produce the same sensitivity and adjoint construction; geometry and constitutive data change its operators. Represent complex controls by their stacked real and imaginary components before taking the real derivatives below.

For these operator derivatives, fields on moving domains are pulled back to fixed reference domains with fixed duality pairings. The residuals then include the coordinate Jacobians, moving-coordinate transport and transformed boundary conditions. An equivalent geometric derivation must retain those same transport and measure-variation terms. Differentiating a fixed-domain equation while silently changing its domain would not give the required shape adjoint.

For the stationary residual $\mathcal{R}$ and objective J_{stat} , introduce an adjoint multiplier $\boldsymbol{\lambda}$ in the dual residual space and the Lagrangian $\mathcal{L} = J_{\text{stat}} + \langle \boldsymbol{\lambda}, \mathcal{R} \rangle$. The superscript $*$ on an operator now denotes its adjoint in the stated real duality pairing, rather than scalar conjugation. Eliminating state variations gives

$$\mathcal{R}_{\mathbf{y}}^* \boldsymbol{\lambda} = -(J_{\text{stat}})_{\mathbf{y}}, \quad \nabla_u j = (J_{\text{stat}})_u + \mathcal{R}_u^* \boldsymbol{\lambda}. \quad (15.4)$$

Here $j(u) = J_{\text{stat}}(\mathbf{y}(u), u)$ is the reduced objective. The derivatives include all state-dependent loads and boundary terms. For the fixed acoustic source subproblem, the plus-sign residual convention uses the negative of the field Z in [section 10](#); both give the same reduced gradient. The adjoint approach supplies many control derivatives from a state solve and an adjoint solve per scalar functional [\[15\]](#). It does not assert convexity or global optimality.

For dynamics, introduce the time-dependent adjoint $\boldsymbol{\Lambda}(t)$ and use

$$\mathcal{L}_T = \Phi_T(\mathbf{y}(T_f)) + \int_0^{T_f} [\ell + \langle \boldsymbol{\Lambda}, \mathcal{D}(\mathbf{y}, \dot{\mathbf{y}}, \mathbf{a}) \rangle] dt. \quad (15.5)$$

Integrating the $\delta \dot{\mathbf{y}}$ term by parts gives the formal adjoint equation and terminal condition

$$\ell_{\mathbf{y}} + \mathcal{D}_{\mathbf{y}}^* \boldsymbol{\Lambda} - \frac{d}{dt}(\mathcal{D}_{\dot{\mathbf{y}}}^* \boldsymbol{\Lambda}) = 0, \quad (15.6)$$

$$(\Phi_T)_{\mathbf{y}} + \mathcal{D}_{\dot{\mathbf{y}}}^* \boldsymbol{\Lambda}(T_f) = 0. \quad (15.7)$$

The terminal equation applies to free terminal state components. Prescribed terminal constraints add multipliers. The explicit example in [section 10](#) supplies those spatial conditions for one wave subproblem. Spatial integrations by parts provide the remaining adjoint boundary and interface conditions; these depend on the actual physical boundaries and objective and cannot be universally omitted. Differential– algebraic constraints require their consistent adjoint treatment or elimination.

For a fixed formation horizon, the interior command gradient is

$$\mathbf{g}_a = \ell_a - \frac{d}{dt}\ell_{\dot{a}} + \mathcal{D}_a^* \mathbf{\Lambda}. \quad (15.8)$$

A free command endpoint has the associated natural boundary term from $\ell_{\dot{a}}$; a prescribed pulse endpoint has no free endpoint variation. If T_f is optimized, its transversality condition must also be included. State and path constraints augment the Lagrangian with their multipliers and complementarity conditions. For a convex admissible history set $\mathfrak{A}_{\text{ad}}$ after those constraints are treated, first-order optimality requires

$$\int_0^{T_f} \langle \mathbf{g}_a, \mathbf{v} - \mathbf{a} \rangle dt \geq 0 \quad \text{for every } \mathbf{v} \in \mathfrak{A}_{\text{ad}}. \quad (15.9)$$

These are necessary local optimality equations under regularity assumptions; they are not a closed-form excitation or a proof of a global optimum.

15.4 Robustness uses one realizable policy

Let $\mathcal{V}$ be a declared set of uncertain parameter values, and let π be a causal policy mapping available measurement histories to commands. A robust formulation is

$$\min_{\pi} \sup_{\boldsymbol{\vartheta} \in \mathcal{V}} J(\mathbf{y}_{\boldsymbol{\vartheta}}, \pi; \boldsymbol{\vartheta}), \quad \text{with the governing and operating constraints for every } \boldsymbol{\vartheta} \in \mathcal{V}. \quad (15.10)$$

The same policy acts in every case; it cannot choose a different command using an unobserved true parameter. Parameter estimation can be part of the observation state. An open-loop pulse is the special case of a policy fixed before observing the response. This separates a repeatable inverse-design method from a shape fitted under a single favorable calibration.

16 Cartesian geometry as a target family

16.1 An unsquared optical-path construction

Fix a vacuum wavelength λ_{opt} , positive unequal homogeneous indices n_o, n_i , and distinct conjugate points $\mathbf{O}, \mathbf{I}$. The signs s_o, s_i are +1 for a real object or real image, respectively, and -1 for a virtual one. For an interface point $\mathbf{x}$ distinct from both foci, let $R_o = |\mathbf{x} - \mathbf{O}|$ and $R_i = |\mathbf{x} - \mathbf{I}|$. Define the signed optical path Ψ and a constant path level C by

$$\Psi(\mathbf{x}) = s_o n_o R_o + s_i n_i R_i, \quad \Gamma_{\text{opt}, \star} \subset \{\mathbf{x} : \Psi(\mathbf{x}) = C\}. \quad (16.1)$$

Both Ψ and C have units of length. The constant is fixed by a chosen vertex or by compatible mounting and volume data. For two real conjugates this is the usual weighted sum of distances. Signed paths extend the construction to virtual conjugates. The formulas and Cartesian-surface classification follow Silva-Lora and Torres [18]; the preceding mechanical and control formulation is separate from that paper’s two-surface singlet construction.

Let $\mathbf{d}_o = s_o(\mathbf{x} - \mathbf{O})/R_o$ be the incident propagation direction and $\mathbf{d}_i = s_i(\mathbf{I} - \mathbf{x})/R_i$ the desired transmitted direction. Then

$$\nabla\Psi = n_o\mathbf{d}_o - n_i\mathbf{d}_i, \quad \nabla_\Gamma\Psi = 0 \implies n_o\mathbf{P}_\Gamma\mathbf{d}_o = n_i\mathbf{P}_\Gamma\mathbf{d}_i. \quad (16.2)$$

This directly supplies the tangential Snell condition. Squaring [equation \(16.1\)](#) gives a quartic, but can add branches; its real roots alone do not certify the intended refracting surface.

To select transmission, let $\mathbf{m}$ be the unit surface normal pointing from the incident medium into the transmitted medium. Thus $\mathbf{m} = \pm\mathbf{n}$ depending on which fluid light enters first. Set $c_o = \mathbf{d}_o \cdot \mathbf{m}$ and $\eta_n = n_o/n_i$. The physically transmitted unit direction $\mathbf{d}_t$ is

$$\mathbf{d}_t = \eta_n\mathbf{d}_o + \left[\sqrt{1 - \eta_n^2(1 - c_o^2)} - \eta_n c_o \right] \mathbf{m}. \quad (16.3)$$

The branch must satisfy $c_o > 0$, a nonnegative square-root argument, $\mathbf{d}_t \cdot \mathbf{m} > 0$, and $\mathbf{d}_t = \mathbf{d}_i$ throughout the admitted aperture. These exclude back-facing and totally internally reflecting branches. A finite distance from grazing transmission is also needed for well-conditioned optical sensitivities.

16.2 The four axial conjugate parameters

Choose the vertex as the origin, with optical propagation there in the positive z direction. Let z_o, z_i be the signed axial coordinates of the conjugates, and define cylindrical radius $r = \sqrt{x^2 + y^2}$. For finite nonzero conjugates, $s_o = -\text{sgn}(z_o)$ and $s_i = \text{sgn}(z_i)$. The level fixed by the vertex is $C = -n_o z_o + n_i z_i$. In particular, for $z_o < 0 < z_i$,

$$n_o\sqrt{r^2 + (z - z_o)^2} + n_i\sqrt{r^2 + (z_i - z)^2} = -n_o z_o + n_i z_i. \quad (16.4)$$

For the paper's parametric representation, define inverse lengths $a = 1/z_o$, $b = 1/z_i$, and $D = n_i a - n_o b$. When the denominators are nonzero, define the form parameters $\mathbf{O}$ (inverse length), $\mathbf{G}$ (dimensionless), $\mathbf{T}$ (inverse length cubed), and $\mathbf{S}$ (inverse length squared) as

$$\mathbf{O} = \frac{n_i b - n_o a}{n_i - n_o}, \quad \mathbf{G} = \frac{(n_i^2 a - n_o^2 b)^2}{n_i n_o D (n_i b - n_o a)}, \quad (16.5)$$

$$\mathbf{T} = \frac{(n_i - n_o)(n_i + n_o)^2 a^2 b^2}{4 n_i n_o D}, \quad \mathbf{S} = \frac{(n_i + n_o)(n_i^2 a - n_o^2 b) a b}{2 n_i n_o D}. \quad (16.6)$$

The signed vertex graph curvature is $z''(0) = \mathbf{O}$; it is not the total mechanical curvature κ . For an upward physical normal at an axisymmetric vertex, $\kappa(0) = -2\mathbf{O}$.

Define the polar distance $\varrho = \sqrt{r^2 + z^2}$, which is *not* the cylindrical radius r or either fluid density. The parameters satisfy $\mathbf{S}^2 = \mathbf{O}\mathbf{G}\mathbf{T}$. The Cartesian polynomial in these coordinates is

$$\mathbf{O}\mathbf{G}z^2 - 2(1 + \mathbf{S}\varrho^2)z + (\mathbf{O} + \mathbf{T}\varrho^2)\varrho^2 = 0. \quad (16.7)$$

Solving this quadratic in z , selecting the branch through the vertex, and rationalizing yields

$$z(\varrho) = \frac{(\mathbf{O} + \mathbf{T}\varrho^2)\varrho^2}{1 + \mathbf{S}\varrho^2 + \sqrt{1 + (2\mathbf{S} - \mathbf{O}^2\mathbf{G})\varrho^2}}, \quad r(\varrho) = \sqrt{\varrho^2 - z(\varrho)^2}. \quad (16.8)$$

The unsquared path and transmission tests remain necessary. Singular GOTS coordinates need not imply a singular physical surface; in that event use [equation \(16.1\)](#) directly. Equal indices, coincident foci, and degenerate conjugates require their own optical interpretation rather than division by a vanishing coefficient.

For collimated incidence, take $z_o \rightarrow -\infty$ and subtract the divergent constant path. The finite condition for a real image is $n_o z + n_i(\sqrt{r^2 + (z_i - z)^2} - z_i) = 0$. Here $\mathbf{S} = \mathbf{T} = 0$ and the conic constant $K = -(n_o/n_i)^2$. Writing the signed vertex curvature as $c_v = n_i/[z_i(n_i - n_o)]$ gives

$$z(r) = \frac{c_v r^2}{1 + \sqrt{1 - (1 + K)c_v^2 r^2}}. \quad (16.9)$$

For finite conjugates, a Cartesian surface is generally not an exact conic, and $\mathbf{G}$ must not be identified with a fitted conic constant.

16.3 A common intermediate conjugate for the two faces

Let the laboratory vertices be v_0, v_1 , the object and final image positions be z_0, z_2 , and the common intermediate axial point be z_1 . With indices n_0, n_1, n_2 , the two local prescriptions are

$$(n_o, z_o, n_i, z_i)_0 = (n_0, z_0 - v_0, n_1, z_1 - v_0), \quad (16.10)$$

$$(n_o, z_o, n_i, z_i)_1 = (n_1, z_1 - v_1, n_2, z_2 - v_1). \quad (16.11)$$

These are the vertex translations of the singlet construction of Silva-Lora and Torres [18, Sec. 3]. Sharing a laboratory point does not mean equating the two vertex-relative distances or merely changing a sign: the vertex separation must also be included. When z_1 lies beyond both faces, the wave between them is converging: the first face has a real image and the second a virtual object at this same point.

For this branch let $\mathbf{x}_0, \mathbf{x}_1$ be successive intersections and let $L_{01} = |\mathbf{x}_1 - \mathbf{x}_0|$. Since the intermediate ray travels toward $\mathbf{Z}_1 = (0, 0, z_1)$ without passing it, $|\mathbf{x}_0 - \mathbf{Z}_1| - |\mathbf{x}_1 - \mathbf{Z}_1| = L_{01}$. Adding the two signed constant-path equations therefore yields

$$n_0|\mathbf{x}_0 - \mathbf{Z}_0| + n_1 L_{01} + n_2|\mathbf{x}_1 - \mathbf{Z}_2| = C_0 + C_1.$$

The intermediate terms cancel into the actual positive propagation length. Together with the physical Snell branches, this constructs an axially stigmatic pair. Other real/virtual placements require their corresponding signed branch and interception checks. Two independently stigmatic surfaces with different intermediate points do not satisfy this construction.

16.4 Geometric compatibility and limits of arbitrariness

For fixed homogeneous indices and two fixed points, Ψ is invariant under rotation about their joining axis. Consequently a regular connected Cartesian level surface belongs to a surface of revolution. An arbitrary non-axisymmetric surface is generally incompatible with exact single-interface point-to-point stigmatism under these optical assumptions. Arbitrary aspheres remain valid mechanical targets, but require a correspondingly different optical objective or incident wavefront.

More generally, let $S_o(\mathbf{x})$ and $S_i(\mathbf{x})$ be incident and desired transmitted eikonal functions, measured as optical path lengths. In homogeneous media they satisfy $|\nabla S_o| = n_o$ and $|\nabla S_i| = n_i$, with gradients pointing along propagation. A purely refracting interface with no imposed phase discontinuity must lie on $S_o - S_i = C$. The point-conjugate choice is $S_o = s_o n_o R_o$, $S_i = -s_i n_i R_i$, recovering [equation \(16.1\)](#). A non-axisymmetric target can thus be coupled to a specified different wavefront-design problem, but the necessary incident wavefront must be physically supplied rather than assumed to arise from the same two point conjugates.

The target must also be compatible with the available fluid mass, supporting boundaries and allowed contact-line motion. In an incompressible model, fixing the entire interface and its other boundaries fixes the enclosed volume. A single optical path constant cannot generally enforce an

arbitrary boundary curve and an independently prescribed volume. Possible design freedoms are a non-optical annulus, a movable support, a reservoir, or revised conjugates; none is silently introduced here.

The incident directions in [equation \(16.2\)](#) are specified in a homogeneous incident medium. If light first crosses a window or another liquid surface, that preceding optical system must produce those directions and optical paths. Similarly, solidifying a liquid can change its index and shape. A liquid Cartesian diopter is therefore not automatically a Cartesian finished lens. Exact axial stigmatism at one wavelength does not establish the Abbe sine condition, off-axis aplanatism, or achromatism.

16.5 Specialization of the general inverse problem

For a Cartesian target this curvature can be evaluated analytically from its unsquared optical path. Define $\mathbf{u}_o = (\mathbf{x} - \mathbf{O})/R_o$, $\mathbf{u}_i = (\mathbf{x} - \mathbf{I})/R_i$, $\mathbf{g}_\Psi = \nabla \Psi$ and the path Hessian $\mathbf{H}_\Psi = \nabla \nabla \Psi$. Choose the orientation sign $\epsilon_n \in \{-1, 1\}$ such that the physical normal is $\mathbf{n} = \epsilon_n \mathbf{g}_\Psi / |\mathbf{g}_\Psi|$. Then

$$\mathbf{H}_\Psi = \frac{s_o n_o}{R_o} (\mathbf{I} - \mathbf{u}_o \otimes \mathbf{u}_o) + \frac{s_i n_i}{R_i} (\mathbf{I} - \mathbf{u}_i \otimes \mathbf{u}_i), \quad (16.12)$$

$$\kappa_C = \frac{\epsilon_n}{|\mathbf{g}_\Psi|} \left(\text{tr } \mathbf{H}_\Psi - \frac{\mathbf{g}_\Psi^T \mathbf{H}_\Psi \mathbf{g}_\Psi}{|\mathbf{g}_\Psi|^2} \right), \quad \Pi_C = \sigma \kappa_C + \Delta \rho \Phi - \lambda_V. \quad (16.13)$$

Thus the four optical parameters determine an explicit required-load field on each selected Cartesian patch once its placement, material pair and body force are given. A non-optical continuation and its boundary conditions remain necessary for the complete fluid problem. This formula avoids differentiating a numerically fitted conic that is only approximately Cartesian.

16.6 From four optical parameters to a conditional source map

Let $\mathbf{p} = (n_o, z_o, n_i, z_i)$ denote the four axial optical parameters. Let $\mathcal{D}$ collect all separately specified physical and operating data: cylinder and optical placement, aperture, wavelength, fluid properties, fill, wall and wetting laws, source supports and response, limits, observations, initial state and required tolerances. The vertex and signed branch conventions above select a Cartesian patch $\Gamma_C(\mathbf{p}; \mathcal{D})$ when it exists. Completing the non-optical surface must also respect $\mathcal{D}$.

Define $\mathfrak{S}(\mathbf{p}; \mathcal{D})$ as the set of source fields and formation/holding policies for which the coupled equations, physical bounds, optical tolerances and specified stability conditions hold for that target. This set can be empty or contain many members. The useful design output is therefore a conditional selection

$$(u_\star, \pi_\star) \in \arg \min_{(u, \pi) \in \mathfrak{S}(\mathbf{p}; \mathcal{D})} J(u, \pi; \mathbf{p}, \mathcal{D}), \quad (16.14)$$

where J is a declared effort, formation and optical objective, and π is a causal holding or formation policy when one is needed. The observation cost J_{obs} in the general formulation can now be a nonnegative combination of J_W , J_d , J_{spot} , with the fast optical corrections just described. The operator equations are reused for each $\mathbf{p}$; their definition does not rely on a neighboring solved conjugate pair.

This notation does not assert that the argmin is attained for the full nonlinear problem. The existence proposition in [section 7](#) applies only to its stated regularized fixed-geometry source subproblem. A device that accepts the four optical numbers must first fix $\mathcal{D}$ and establish which parameter region admits an acceptable result. It should report a demonstrated infeasibility condition or an unresolved solve when appropriate, rather than label the nearest available surface an exact oval. Global phase and other source nonuniqueness also require a stated selection

convention. These issues concern a general, reproducible inverse method; continuation remains an optional tool for investigating its solutions.

17 The two-face ten-nanometre specification

Each Cartesian target is fixed in laboratory coordinates, with a declared vertex, clear aperture and compatible non-optical continuation. The two targets may be chosen independently as geometries; two arbitrary Cartesian dioptrics do not automatically compose an aplanatic or stigmatic singlet. The present objective is geometrical shaping. The optical quantities in [section 16](#) provide optional downstream diagnostics.

For mean graph surfaces $\bar{h}_j$ on apertures $D_{o,j}$, define

$$e_{\text{mean}}(t) = \max_{j=f,b} \sup_{D_{o,j}} |\bar{h}_j - h_{C,j}|, \quad \epsilon = 10^{-8} \text{ m.} \quad (17.1)$$

No piston, tilt, best-fit sphere, or refocusing is removed. RMS error cannot replace this maximum. The spatial search must include interior extrema between nodes. A sampled maximum is a lower bound on the continuum supremum until an interpolation or derivative bound encloses the omitted points.

With a single carrier and first-order normal displacement Ξ_j , the instantaneous graph is $h_j = \bar{h}_j + \text{Re}[(\Xi_j/n_{z,j})e^{-i\omega t}] + O(|\Xi|^2)$. At frozen slow time, maximization over the carrier phase and the aperture gives

$$e_{\text{phase}} = \max_{j=f,b} \sup_{D_{o,j}} \left(|\bar{h}_j - h_{C,j}| + \frac{|\Xi_j|}{|n_{z,j}|} \right). \quad (17.2)$$

This is exact for the retained single-harmonic graph model. Multiple carriers, beat motion and nonlinear oscillations need their actual time-dependent sum or a justified bound. Excluding optical index modulation from the objective does not exclude physical motion of the surface.

Acceptance over a declared holding interval $[T_f, T_h]$ requires a justified upper bound on the selected physical maximum throughout that interval. An error budget must account for mean fit, numerical error, uncertainty, and disturbances without treating unknown contributions as zero. Fast motion is added only for the stricter instantaneous specification; its indirect effects on the mean remain model-validity questions. Fixed-command refinement supplies a numerical error estimate, not automatically a rigorous upper bound. Model omission bounds require independent physics or measurements. Formation starts at the solved unforced equilibrium with sources off; a flat surface is a permissible initial condition only when it solves that problem. Stored energy, fluid velocity and thermal state must also be initialized.

For both faces, the slow linearized block contains

$$\mathbf{K}_{\text{eff}} = \mathbf{K}_{\text{cap+grav}} - \begin{pmatrix} D_{hf}q_f & D_{hb}q_f \\ D_{hf}q_b & D_{hb}q_b \end{pmatrix}.$$

It is projected using the *actual* volume constraints, coupled to the fluid mass/mobility and linearized about the actual mean flow. The angular sectors and memory requirements of [section 11](#) apply to this block system. Thus reaching a small static residual, maintaining a cycle-mean shape, and maintaining the instantaneous physical tolerance are separately testable scientific milestones.

Selected specification, 23 September 2026. The user-selected objective is the *cycle-mean driven-liquid surface*, not the instantaneous interface and not a cured solid. Thus the active geometric gate is $e_{\text{mean}} \leq \epsilon$ on both apertures. Carrier amplitude is not added to this error

and need not itself be below ten nanometres. Equation (17.2) remains a separate, stricter diagnostic, as do the earlier carrier-aware experiments. The mean specification still needs a holding-interval error budget, numerical convergence and mean-flow stability. Radiation traction, nonlinear averaging, streaming and thermal feedback can change the cycle mean; removing the instantaneous-motion penalty does not justify neglecting those mechanisms.

18 Carrier-aware numerical inverse and acceptance gates

At a prescribed geometry, the finite-source operators supply generalized loads $F_i(\mathbf{a}) = \mathbf{a}^\dagger \mathbf{K}_i \mathbf{a}$ and sampled graph-displacement amplitudes $b_s(\mathbf{a}) = \mathbf{B}_s \mathbf{a}$, with $\mathbf{B}_s = \mathbf{V}_s / (\omega |n_{z,s}|)$. Both maps arise from the same chamber solution. The matrices $\mathbf{B}_s$ have units of seconds when source commands have units of velocity. With the actual volume constraints already imposed, let $\mathbf{K}_m$ be the capillary-gravity Hessian and $\mathbf{O}$ evaluate a surface perturbation on the fixed apertures. The surrogate

$$\mathbf{d}(\mathbf{a}) = \mathbf{O} \mathbf{K}_m^{-1} [\mathbf{F}(\mathbf{a}) - \mathbf{F}_{\text{required}}] \quad (18.1)$$

has units of length but is *not* the coupled equilibrium error. Its inverse stiffness omits the shape derivative of the acoustic load. A small surrogate residual cannot replace a fresh solution of the full stationary equations, much less a stability calculation.

For the selected cycle-mean specification, the carrier penalty may be set to zero while retaining the full quadratic acoustic load and the same source constraints. Carrier-aware constructions below are optional stricter studies, not mandatory gates for mean geometry. A small frozen mean residual still requires the same independent coupled forward and fixed-command convergence tests; changing the metric does not convert a surrogate into an achieved shape.

18.1 Coherent source steps with explicit carrier constraints

A source-step model at $\mathbf{a}_k$ uses the exact directional derivative

$$DF_i(\mathbf{a}_k)[\delta \mathbf{a}] = 2 \operatorname{Re}\{\mathbf{a}_k^\dagger \mathbf{K}_i \delta \mathbf{a}\}. \quad (18.2)$$

The carrier trace is already linear. Thus one may minimize an epigraph variable t subject to convex constraints on the linearized surrogate, the exact carrier trace and the physical source-amplitude disks:

$$|d_s(\mathbf{a}_k) + Dd_s(\mathbf{a}_k)[\delta \mathbf{a}]| \leq u_j, \quad s \text{ on face } j, \quad (18.3)$$

$$|\mathbf{B}_s(\mathbf{a}_k + \delta \mathbf{a})| \leq v_j, \quad s \text{ on face } j, \quad (18.4)$$

$$u_j + v_j \leq t, \quad |a_{k,\ell} + \delta a_\ell| \leq a_{\ell,\max}. \quad (18.5)$$

A trust region bounds the source step; a separately declared annulus penalty can control the omitted non-optical region. The sum of separate facewise maxima is conservative relative to the maximum of the pointwise sum, but both remain sampling diagnostics. Each proposed step is checked against the *true quadratic* frozen-geometry load before acceptance. The apparatus, targets and frequency stay fixed during these iterations: this is neither a physical trajectory nor target continuation. A numerical termination flag is reported separately from first-order stationarity, feasibility and attainment of the geometrical criterion.

Source-coordinate conditioning may use the eigenspaces of $\mathbf{G} = \sum_i \mathbf{K}_i^\dagger \mathbf{K}_i$. Any eigenvalue truncation restricts the admissible numerical source space and must be declared. It does not establish the physical rank of the continuous source operator. More emitting regions can introduce nearly dependent controls rather than improve reachability.

18.2 Sampled peak majorants

An alternative smooth-away-from-zero source objective uses unnormalized finite-vector norms on each aperture. With $\epsilon = 10^{-8}$ m, define $\mathbf{h}_j = \mathbf{d}_j/\epsilon$, $\mathbf{c}_j = \mathbf{b}_j/\epsilon$, and

$$s_j = \|\mathbf{h}_j\|_p + \|\mathbf{c}_j\|_p, \quad p \geq 2. \quad (18.6)$$

Then $\epsilon s_j \geq \max_s |d_{j,s}| + \max_s |b_{j,s}|$ on the declared sample sets, even if the two sets differ. No division by the sample count is made: such a normalization would destroy this majorant. The implemented least-squares objective uses the two residuals s_j together with separate annulus and source-effort penalties. This is not exact minimax optimization. For a nonzero complex vector,

$$D\|\mathbf{v}\|_p[\delta\mathbf{v}] = \frac{\sum_k |v_k|^{p-2} \operatorname{Re}(\overline{v_k} \delta v_k)}{\|\mathbf{v}\|_p^{p-1}}. \quad (18.7)$$

At the all-zero vector a zero subgradient is selected. Directional numerical checks cover both physical and reduced source coordinates. Neither this majorant nor a large choice of p controls unsampled extrema or repairs the omitted acoustic shape derivative in $\mathbf{d}$.

A complementary, deliberately conservative source restriction chooses an orthonormal matrix $\mathbf{N}$ with $\|\mathbf{BN}\|_2 \leq \epsilon_a/(\sqrt{N_s}a_{\max})$. Commands in its range satisfying $|a_\ell| \leq a_{\max}$ obey

$$\|\mathbf{Ba}\|_\infty \leq \|\mathbf{BN}\|_2 \|\mathbf{a}\|_2 \leq \epsilon_a. \quad (18.8)$$

The operator norm is checked after numerical eigenspace selection. This is only a sampled frozen-model carrier bound; failure to match the required traction in this restricted subspace is not a reachability result for the full source space.

18.3 A relaxation diagnostic, not a substitute waveform

In the lifted variable of [equation \(8.5\)](#), a sampled carrier bound is linear:

$$\operatorname{tr}(\mathbf{B}_s^\dagger \mathbf{B}_s \mathbf{W}) \leq \epsilon_a^2. \quad (18.9)$$

The corresponding minimum-carrier semidefinite relaxation can help distinguish an unsuccessful coherent inverse from a source-space limitation. A higher-rank covariance is not a coherent command. Inaccurate numerical solutions that violate positivity, traction equalities or source bounds supply neither a feasible design nor a verified lower-bound certificate. Bounds derived in a truncated source space cannot be extrapolated to the full continuous space.

18.4 Why the maximum spot also requires slope control

For fixed indices, write the physical Snell map as $\mathbf{d}_t = \eta_n \mathbf{d} + (\sqrt{Q} - \eta_n c) \mathbf{m}$, where $c = \mathbf{d} \cdot \mathbf{m}$ and $Q = 1 - \eta_n^2(1 - c^2) > 0$. Its first variation is

$$\delta \mathbf{d}_t = \eta_n \delta \mathbf{d} + \left(\frac{\eta_n^2 c}{\sqrt{Q}} - \eta_n \right) \delta c \mathbf{m} + (\sqrt{Q} - \eta_n c) \delta \mathbf{m}, \quad (18.10)$$

$$\delta c = \delta \mathbf{d} \cdot \mathbf{m} + \mathbf{d} \cdot \delta \mathbf{m}. \quad (18.11)$$

At a fixed horizontal coordinate of a graph, its upward normal satisfies

$$\delta \mathbf{m} = \frac{(\mathbf{I} - \mathbf{m} \mathbf{m}^T)(-\nabla \delta h, 0)}{\sqrt{1 + |\nabla h|^2}}.$$

Evaluating the normal at a moving intersection adds the change of its evaluation point. Both effects, and the preceding interface's change to $\mathbf{d}$, must be included in a complete ray derivative.

In particular, small height error without a slope bound does not bound the outgoing angle; the derivative also becomes ill-conditioned near $Q = 0$.

For propagation from $\mathbf{x}$ to a fixed detector plane $z = Z$, let $\tau = (Z - x_z)/d_z > 0$ and $\mathbf{y}_\perp = \mathbf{x}_\perp + \tau \mathbf{d}_\perp$. Then

$$\delta\tau = -\frac{\delta x_z + \tau \delta d_z}{d_z}, \quad \delta \mathbf{y}_\perp = \delta \mathbf{x}_\perp + \tau \delta \mathbf{d}_\perp + \mathbf{d}_\perp \delta\tau.$$

These expressions explain the amplification of short-scale surface errors into detector displacement. They do not approximate a diffraction image.

A declared regularity proxy augments the frozen height observations with

$$\mathbf{d}_s(\mathbf{a}) = \ell \mathbf{D}_r \mathbf{K}_m^{-1} [\mathbf{F}(\mathbf{a}) - \mathbf{F}_{\text{required}}], \quad [\ell] = m.$$

Here $\mathbf{D}_r$ evaluates pupil slopes and $\mathbf{d}_s$ again has units of length. For $\ell = 10^{-3}$ m, a 10^{-5} slope error contributes like a 10^{-8} m height error. The already-derived source gradient is composed with this additional linear observation. It does not remove gravity from $\mathbf{K}_m$ or $\mathbf{F}_{\text{required}}$. Unnormalized p -norms may majorize sampled height and scaled-slope peaks; for a mean-only specification the carrier contribution is explicitly omitted, while its diagnostic is retained. Neither the slope scale nor this majorant is an exact optical bound.

The independent joint gate requires each clear-pupil mean height error to be at most 10^{-8} m and the maximum geometric radius about the prescribed optical axis at the fixed detector to be at most 10^{-6} m. RMS radius, centroid recentering, refocusing, and discarding failed rays cannot substitute for this maximum. The intended illumination and pupils must be declared and checked for full transmission. Discrete ray samples must be refined and distinguished from a continuous-pupil certificate; these tests still do not establish maintained liquid accuracy.

18.5 Static acoustic feedback in the precision inverse

The source conditions above must not confuse the mechanical compliance with the coupled static response. At a fixed reference shape and command, write

$$\mathbf{R}(\mathbf{q}, \mathbf{a}) = \mathbf{F}_m(\mathbf{q}) - \mathbf{F}_a(\mathbf{q}, \mathbf{a}), \quad \mathbf{A}_q = D_q \mathbf{F}_a, \quad \mathbf{J} = \mathbf{K}_m - \mathbf{A}_q.$$

The first-order equilibrium correction solves $\mathbf{J} \delta \mathbf{q} = -\mathbf{R}$. If $\mathbf{J}$ is invertible, the local equilibrium source derivative is $D_a \mathbf{q} = \mathbf{J}^{-1} D_a \mathbf{F}_a$. An observation of the actual coupled response therefore contains $\mathbf{O} \mathbf{J}^{-1}$, and the fixed-detector optical observation contains $\mathbf{J}_{\text{spot}} \mathbf{J}^{-1}$. The latter includes both surfaces, not separate single-face focus errors. A small $\mathbf{O} \mathbf{K}_m^{-1} \mathbf{R}$ need not imply a small $\mathbf{O} \mathbf{J}^{-1} \mathbf{R}$: the acoustic shape feedback can strongly amplify some force defects, including those in the non-optical annuli.

As an explicitly truncated diagnostic, choose admissible shape directions in the columns of $\mathbf{V}$ and, for positive definite $\mathbf{K}_m$, define

$$\mathbf{W} = (\mathbf{V}^T \mathbf{K}_m \mathbf{V})^{-1} \mathbf{V}^T \mathbf{K}_m, \quad \mathbf{L} = \mathbf{K}_m^{-1} \mathbf{A}_q \mathbf{V}.$$

Replacing $\mathbf{A}_q$ by $(\mathbf{A}_q \mathbf{V}) \mathbf{W}$ gives

$$\mathbf{C}_V = \mathbf{K}_m^{-1} + \mathbf{L}(\mathbf{I} - \mathbf{W} \mathbf{L})^{-1} \mathbf{W} \mathbf{K}_m^{-1}. \quad (18.12)$$

Direct multiplication verifies this inverse whenever the small matrix is nonsingular. Central differences of newly solved wave fields can measure $\mathbf{A}_q \mathbf{V}$; the step and retained directions must be refined. This is a restriction of the general coupled linearization, not a replacement for it or for the continuous-source inverse conditions.

Holding $\mathbf{C}_V$ fixed during a source step supplies only a local model. The acoustic derivative itself depends on the command, so finite source changes require fresh coupled validation and, when needed, an updated response operator. Rank-truncated static conditioning is neither a stability spectrum nor an infeasibility certificate. In particular it does not test three-dimensional disturbances, inertia, dissipative dynamics, a streaming base flow or thermal feedback.

18.6 Required independent verification

For any candidate, solve the nonlinear coupled equilibrium with fixed physical commands and refine both wave and surface spaces. The absolute mean error and carrier amplitude must be evaluated independently of design collocation. Passing a sampled ten-nanometre check still requires a numerical-error budget and control of unsampled extrema. Formation, stability about the appropriate streaming base flow, thermal feedback, material uncertainty and experimental validation remain additional gates. The numerical inverse does not remove those requirements by using more emitters or higher carrier frequencies.

19 Regimes, unresolved physics and the route to a design

19.1 Scales that determine which equations may be reduced

Choose a characteristic geometric length $L > 0$, slow velocity U_s , slow variation time T_s , acoustic velocity amplitude V_a , sound speed c , and reference density, viscosity and surface tension ρ, μ, σ . These are representative positive scales, not assigned apparatus values. Let $\nu = \mu/\rho$ be kinematic viscosity, D_T thermal diffusivity, and Π_a a characteristic acoustic traction. Where gravity is relevant, g_{grav} is its magnitude, distinct from the boundary source g . Define the following dimensionless ratios and times:

$$\text{Ma}_a = V_a/c, \quad \text{Re}_s = \rho U_s L / \mu, \quad \mathcal{I}_s = \rho L^2 / (\mu T_s), \quad (19.1)$$

$$\text{Oh} = \mu / \sqrt{\rho \sigma L}, \quad \text{Bo} = |\Delta \rho| g_{\text{grav}} L^2 / \sigma, \quad \text{Ca}_a = \Pi_a L / \sigma, \quad (19.2)$$

$$\tau_\sigma = \sqrt{\rho L^3 / \sigma}, \quad \tau_\mu = \mu L / \sigma, \quad \tau_\nu = \rho L^2 / \mu, \quad (19.3)$$

$$\delta_\nu = \sqrt{2\nu/\omega}, \quad \delta_T = \sqrt{2D_T/\omega}. \quad (19.4)$$

Here $\mathcal{I}_s$ measures unsteady inertia relative to viscosity, Oh is the Ohnesorge number, Bo the Bond number, and Ca_a an acoustic traction/capillary scale ratio. The three τ quantities are capillary–inertial, viscous–capillary and momentum–diffusion times; δ_ν, δ_T are viscous and thermal penetration depths. Both fluids may supply important scales, so a favorable ratio in one phase does not justify eliminating the other.

Reduction	Conditions to examine before using it
Geometric optics	Optical wavelength small relative to relevant curvature and wave-front scales; intended polarization, dispersion and diffraction requirements treated separately.
Linear acoustics	$\text{Ma}_a \ll 1$, small fractional thermodynamic perturbations, and acoustic particle displacement small relative to local geometric scales. The eventual mean interface deformation need not be small.
Thin acoustic layers	δ_ν, δ_T small compared with relevant wall curvature, tangential variation and cavity scales for the particular boundary-layer approximation [9, 4].

Reduction	Conditions to examine before using it
Cycle averaging	A window satisfying equation (4.1) ; beats, modulation and optical exposure must also be resolved or justified as negligible.
Settled harmonic field	Acoustic ring-down short relative to envelope and interface changes, with compatible actuator settling.
Creeping mean flow	Both $\text{Re}_s \ll 1$ and $\mathcal{I}_s \ll 1$ over the formation and perturbation times being studied. Small convective inertia alone does not remove capillary oscillations.
Hydrostatic target load	Negligible mean flow, compatible tangential balance, constant density and surface tension, and the declared volume and wetting law.
Isothermal optics	Thermally induced shape, wave-transfer and index changes small compared with the requested tolerances over the full operating time.

As an explicitly limited analytic example, consider two inviscid, infinitely deep fluids separated by a flat horizontal interface, with the denser fluid below. Let $k_\perp > 0$ be an interfacial spatial wavenumber and $\omega_\Gamma(k_\perp)$ its gravity–capillary angular frequency. Then

$$\omega_\Gamma^2(k_\perp) = \frac{\Delta\rho g_{\text{grav}} k_\perp + \sigma k_\perp^3}{\rho_- + \rho_+}. \quad (19.5)$$

This is a scale check for fast interface compliance, not the eigenfrequency of an unspecified chamber. Finite depth, wetting, curvature and viscosity change its response; the viscous problem has a nontrivial dispersion relation [16]. In particular, making the carrier much faster than the relevant interfacial modes is one ingredient in neglecting first-order capillary corrections to acoustic scattering. The actual coupled boundary problem determines whether that neglect is accurate enough optically.

19.2 Physical effects that may set the attainable optical error

No single hierarchy is guaranteed for all materials and arrangements. The following effects enter the same governing problem when their consequences are comparable with the requested optical tolerance:

1. *Streaming and absorption*: steady recirculation changes viscous tractions, temperature and the acoustic field. It is not equivalent to adding a fitted normal-pressure offset.
2. *Thermal and composition feedback*: density, viscosity, sound speed, attenuation, surface tension and optical index can all depend on temperature or composition. Surface-tension gradients create additional tangential stress.
3. *Fast optical modulation*: interface motion and index modulation in [equation \(B.4\)](#) may affect an exposure even when the mean shape is exactly Cartesian.
4. *Wetting and rheology*: hysteresis, evaporation, contamination, non-Newtonian response or curing alter the state variables or the admissible equilibria. Each requires its own measured or justified closure.
5. *Actuator and boundary loading*: radiation, elastic compliance, transducer coupling and source calibration affect both attainable loads and their shape derivatives. A finite array is not an arbitrary pressure boundary.
6. *Disturbances and resolution*: vibration, parameter drift and finite sensor bandwidth act on the actual dynamic system. Tolerances must account for their optical effect, not only a deterministic design residual.

These are model-selection questions, not reasons to include every effect indiscriminately. The decision is made by scales and sensitivity against a stated optical requirement.

Finite actuator authority particularly limits arbitrary fine spatial detail. For a nearly flat surface with a small sinusoidal height component of wavenumber $k_{\perp}$, the capillary part of the required traction scales as $\sigma k_{\perp}^2$ times its height amplitude. Thus fine features require large curvature loads. Whether the available wave fields can provide those loads is determined by [equation \(8.3\)](#), not by the number of pixels in a target image. Cartesian surfaces form a restricted optical family, which is a useful design focus, but their membership in that family does not prove actuatability.

19.3 A theoretical specification before any numerical experiment

The general problem is specified in this order:

1. Supply the cylinder, fluid laws, conserved mass or volume, initial state, wall response and wetting conditions. Choose a class of regular admissible surfaces without imposing source or target axisymmetry.
2. Specify the physical source supports, commanded variables, constitutive source response, amplitude, bandwidth and energy limits. Select justified wave, mean-flow and thermal models and their averaging regime.
3. For a target surface, impose its geometric and volume compatibility. Derive the required normal load when the quiescent reduction applies; otherwise retain the full stationary flow and stress equations.
4. Formulate the continuous source inverse, test necessary load bounds and derive its state, adjoint and constrained optimality equations. Define the physical tolerance and regularization independently of numerical resolution.
5. Linearize the actual maintained state, including memory and all admitted three-dimensional disturbances. Determine the required observations and source authority for feedback, if the open-loop state is unstable.
6. Specify formation from the supplied initial state and the intended terminal condition: driven operation, temporary transient use, or a separately modeled retention mechanism. State uncertainty and the one realizable control policy.
7. Specialize the target and observations to Cartesian optics if desired. The conjugates select an optical surface; they do not replace the preceding physical data or feasibility conditions.

This construction supplies a reusable inverse formulation. A finite array, a numerical basis, and a nonlinear solution algorithm are later choices. Continuation can explore branches and expose folds; it is not the definition of the inverse map, a required source-generation procedure, or evidence of coverage of the target family.

19.4 Numerical experiments will test stated predictions

The following tests are consequences of the equations, to be performed when numerical work resumes. None is reported as executed in this revision.

Prediction or identity	Experiment and what a discrepancy would mean
Source work, equation (5.10)	Compare integrated source power with boundary absorption using the same solution and units. A persistent difference indicates a sign, phasor normalization, boundary, or omitted-power-term error.
Source adjoint, equations (10.10) and (10.13)	Check the Green identity and independent directional derivatives for boundary and volume sources. Include a velocity-dependent interface objective; a pressure-only check cannot detect a missing adjoint pressure jump.
Shape sensitivity, equations (7.15) and (11.4)	Perturb geometry while holding the same actual physical command fixed. Update wave scattering, normals, measure, wall loading and mean flow. Compare the complete directional derivative under separate refinements.
Angular selection, section 5	In a rotationally invariant geometry, excite specified angular wave orders and check the predicted load differences $m - n$. Test asymmetric geometry separately, where modes should couple.
Source authority, equation (7.5)	Compare optimized loads with independent lower bounds on source effort and with relaxation bounds. A violating solution signals an operator, normalization or constraint error. Failure of an optimizer alone does not prove infeasibility.
Passive dynamics, equation (A.4)	Switch off actuation in a closed passive limit. Check mass and dissipation. A non-capillary stationary shape with every sustaining effect removed contradicts the retention condition.
Coupled stability, equation (11.3)	Perturb the complete holding state, including non-axisymmetric modes, thermal and wave memory when retained. Compare growth with the linearized prediction inside its local range.
Source-space approximation	Refine an admissible source basis with the physical model fixed. Separately refine wave, interface, flow and time discretizations. Agreement is evidence for those instances, not a proof of the compactness hypothesis or the general approximation proposition.
Cartesian specialization	Check the unsquared path, the transmitted Snell branch, analytic curvature and actual ray errors. Include fast interface and index modulation if they exceed the optical tolerance.

Earlier apparatus-specific calculations remain examples under their declared approximations. They do not set the source space or prove the propositions in this manuscript. Before interpreting re-optimization on a finer model, establish convergence with the same physical drive and account for every material loss mechanism. Experimental validation subsequently needs calibrated source motion, material properties, measured interface dynamics and optical response.

19.5 What this formulation establishes and leaves open

The source-to-wave-to-load map, conditional reachability bounds, regularized existence argument, explicit acoustic source adjoint, and constrained dynamic optimality equations provide a common theoretical entry to inverse shaping. Cartesian surfaces supply a useful target family inside that problem. No universal excitation formula follows from four optical parameters alone.

The next theoretical specialization must choose constitutive and wall laws accurate enough for an intended physical experiment, derive the remaining thermoviscous and moving-interface adjoint terms, and establish the relevant regularity and stability bounds. Numerics can test approximations and proposed algorithms, including their failure cases; they cannot promote

assumptions about unmodeled physics into a theorem. A stable machine covering a stated Cartesian parameter region ultimately requires a documented feasible region, source selection rule, formation policy and validation throughout that region.

A Derivations and consistency checks

A.1 Optical path, branch selection and Cartesian curvature

For an arbitrary virtual tangential displacement $\boldsymbol{\tau}_t$ at a surface point, stationarity of the signed optical path reads $\delta\Psi = (n_o\mathbf{d}_o - n_i\mathbf{d}_i) \cdot \boldsymbol{\tau}_t = 0$. This proves [equation \(16.2\)](#). Conversely, on a connected regular patch, tangential Snell refraction into the prescribed point implies $\nabla_\Gamma\Psi = 0$ and hence a constant signed path. The equality of tangential components does not select the direction of the transmitted normal component, which is why the branch tests accompanying [equation \(16.3\)](#) are indispensable.

For a nonzero level gradient $\mathbf{g}_\Psi$, differentiating its unit direction gives

$$\nabla \cdot \frac{\mathbf{g}_\Psi}{|\mathbf{g}_\Psi|} = \frac{\text{tr } \mathbf{H}_\Psi}{|\mathbf{g}_\Psi|} - \frac{\mathbf{g}_\Psi^T \mathbf{H}_\Psi \mathbf{g}_\Psi}{|\mathbf{g}_\Psi|^3}. \quad (\text{A.1})$$

The Hessian of a distance from a point is $(\mathbf{I} - \mathbf{u} \otimes \mathbf{u})/R$, where R is that distance and $\mathbf{u}$ its radial unit vector. These two identities establish [equation \(16.13\)](#) without a paraxial approximation.

The rational Cartesian parameterization follows from solving [equation \(16.7\)](#) as a quadratic in z . Its discriminant is

$$(1 + \mathbf{S}\varrho^2)^2 - \text{OG}(\mathbf{O} + \mathbf{T}\varrho^2)\varrho^2 = 1 + (2\mathbf{S} - \mathbf{O}^2\mathbf{G})\varrho^2, \quad (\text{A.2})$$

because $\mathbf{S}^2 = \text{OGT}$. The root vanishing at the vertex is rationalized to obtain [equation \(16.8\)](#). This cancellation does not excuse skipping the original unsquared path equation.

A.2 Area, curvature and the pressure multiplier

Define a dimensionless perturbation parameter ε and a displaced surface by $\mathbf{x}_\varepsilon = \mathbf{x} + \varepsilon\boldsymbol{\eta}\mathbf{n}$. A dot in this subsection denotes differentiation with respect to ε at zero. The elementary geometric variations are

$$\dot{\mathbf{n}} = -\nabla_\Gamma\boldsymbol{\eta}, \quad (\dot{\text{d}}A) = \kappa\boldsymbol{\eta}\text{d}A, \quad \dot{\kappa} = -\Delta_\Gamma\boldsymbol{\eta} - |\mathbf{B}_\Gamma|^2\boldsymbol{\eta}. \quad (\text{A.3})$$

For a sphere, $\kappa = 2/R_s$ and a uniform radial displacement gives $\dot{\kappa} = -2\boldsymbol{\eta}/R_s^2$, in agreement with direct differentiation. Domain transport gives $\delta\int_{\Omega^-}\Phi\text{d}V = \int_\Gamma\Phi\boldsymbol{\eta}\text{d}A$. Using the functional $E - \lambda_V(V - V_0)$ with acoustic virtual work then yields [equation \(2.10\)](#). At equilibrium, varying its residual gives [equation \(11.5\)](#), including $-\delta\lambda_V$ before projection. Derivative terms from the surface measure multiply the vanishing residual; they cannot be discarded indiscriminately away from equilibrium.

A.3 Dissipation after actuation has ended

Consider the restricted incompressible, constant-property, unforced problem with fixed outer walls, no incoming flux, continuous interfacial velocity and either a pinned contact line or a consistent dissipative wetting/slip law. Define total kinetic energy $K_f = \sum_{\ell \in \{-, +\}} \int_{\Omega^\ell} \rho_\ell |\mathbf{U}_\ell|^2/2 \text{d}V$. Let $\mathcal{D}_{\text{wall}} \geq 0$ denote wall-slip and contact-line dissipation; for the line law [equation \(2.7\)](#), its line contribution is $\int_{\mathcal{C}} \zeta_{\mathcal{C}} V_{\mathcal{C}}^2 \text{d}s$. Multiplying momentum by velocity, integrating by parts and using the interface balance gives

$$\frac{\text{d}}{\text{d}t}(K_f + E) = - \sum_{\ell} \int_{\Omega^\ell} 2\mu_\ell \mathbf{D}(\mathbf{U}_\ell) : \mathbf{D}(\mathbf{U}_\ell) \text{d}V - \mathcal{D}_{\text{wall}} \leq 0. \quad (\text{A.4})$$

The surface work cancels the rate of capillary and wetting energy, and gravity cancels the rate of gravitational energy. The identity provides a check on a future flow solver and explains the loss of pulse energy. It does not prove global convergence, exclude metastable equilibria, or apply unchanged during heating, mass exchange, curing or ongoing wave forcing.

A.4 Quadratic sensitivities and local load authority

At fixed geometry define the real vector $\mathbf{w}_R = (\text{Re } \mathbf{w}, \text{Im } \mathbf{w})^T \in \mathbb{R}^{2N}$. For a Hermitian $\mathbf{K}_i$, variation of the generalized force is

$$\delta F_i = 2 \text{Re}[(\mathbf{K}_i \mathbf{w})^\dagger \delta \mathbf{w}], \quad (\mathbf{J}_F)_{i,:} = 2(\text{Re}(\mathbf{K}_i \mathbf{w})^T, \text{Im}(\mathbf{K}_i \mathbf{w})^T), \quad (\text{A.5})$$

where $\mathbf{J}_F$ is the real load Jacobian. For a nonzero drive, a common phase shift has tangent $\delta \mathbf{w} = i\mathbf{w} \delta \varphi$ with real phase increment $\delta \varphi$, and leaves every F_i unchanged. Thus

$$\text{rank } \mathbf{J}_F \leq \min(M, 2N - 1). \quad (\text{A.6})$$

Consequently a finite coherent array cannot independently prescribe an unlimited number of infinitesimal load components at a fixed geometry. At zero drive the first derivative vanishes because the load is quadratic; finite actuation can still deform the surface. Bounds further restrict the available directional changes.

For a family of targets, changes in Γ also change $\mathbf{K}_i$ and b_i . Linearizing the target-load equations gives

$$(\mathbf{J}_F \delta \mathbf{w}_R)_i = \delta b_i - \mathbf{w}^\dagger (\delta \mathbf{K}_i) \mathbf{w}. \quad (\text{A.7})$$

The right side includes the complete curvature, wave and surface-measure derivatives. It must lie in the admissible Jacobian range for that infinitesimal target change. This identity is a local condition, not a requirement to construct targets by continuation. Small singular values after physical scaling imply sensitive control demands. This is a local test of load authority, distinct from dynamic controllability and from a global existence result. The four-parameter Cartesian family is restricted enough that the dimensional bound alone need not obstruct it; range compatibility, limits and stability still matter.

A.5 Discrete moving-graph acoustic tangent

For the axisymmetric numerical specialization only, use physical reference coordinates (r, ζ) and a layer map $(r, \zeta) \mapsto (r, z(r, \zeta; \mathbf{q}))$. Define $s = \partial_r z$ and $\tau = \partial_\zeta z > 0$. At fixed pulled-back pressure and test functions, the volume part of the wave form is

$$a_q(P, v) = \sum_\ell \int_{\hat{\Omega}_\ell} \frac{2\pi r}{\rho_\ell} \left[\tau P_r \bar{v}_r - s(P_r \bar{v}_\zeta + P_\zeta \bar{v}_r) + \frac{1+s^2}{\tau} P_\zeta \bar{v}_\zeta - k_\ell^2 \tau P \bar{v} \right] dr d\zeta, \quad k_\ell = \omega/c_\ell. \quad (\text{A.8})$$

This follows directly from the inverse map and its determinant, not from a frozen physical gradient. With $\delta s = \partial_r \delta z$ and $\delta \tau = \partial_\zeta \delta z$, its directional derivative is

$$\delta a_q(P, v) = \sum_\ell \int_{\hat{\Omega}_\ell} \frac{2\pi r}{\rho_\ell} \left\{ \delta \tau \left[P_r \bar{v}_r - \frac{1+s^2}{\tau^2} P_\zeta \bar{v}_\zeta - k_\ell^2 P \bar{v} \right] + \delta s \left[-P_r \bar{v}_\zeta - P_\zeta \bar{v}_r + \frac{2s}{\tau} P_\zeta \bar{v}_\zeta \right] \right\} dr d\zeta. \quad (\text{A.9})$$

When the map is the identity on the rigid source boundaries, the supports, admittances, material constants and commands are fixed, their form and load have zero shape derivative. The assembled system then gives

$$\mathbf{A} \delta \mathbf{p} = -(\delta \mathbf{A}) \mathbf{p}.$$

If those boundary assumptions change, their derivatives must be restored. The same wave factorization can serve multiple shape right-hand sides; one must not replace this derivative by a matrix formed at another geometry.

For the common normal-velocity weak trace, write

$$\mathbf{M}_\Gamma \mathbf{u}_n = \frac{\mathbf{A}_{\leq j} \mathbf{p} - \mathbf{f}_{\leq j}}{i\omega}.$$

Here the prefix-domain operator includes its exterior boundary terms and uses the upward interface normal. Consequently

$$\delta \mathbf{u}_n = \mathbf{M}_\Gamma^{-1} \left[\frac{(\delta \mathbf{A}_{\leq j}) \mathbf{p} + \mathbf{A}_{\leq j} \delta \mathbf{p}}{i\omega} - (\delta \mathbf{M}_\Gamma) \mathbf{u}_n \right].$$

For a graph $h(r)$, the mass-measure derivative contains $h' \delta h' / \sqrt{1 + h'^2}$, while its tangential pressure derivative obeys

$$g_t = \frac{\partial_r P_\Gamma}{\sqrt{1 + h'^2}}, \quad \delta g_t = \frac{\partial_r \delta P_\Gamma}{\sqrt{1 + h'^2}} - g_t \frac{h' \delta h'}{1 + h'^2}.$$

At fixed reference interface coordinates, δP_Γ is the trace of $\delta \mathbf{p}$. These geometric trace terms cannot be dropped.

In this transmission model, put

$$c_p = \frac{1}{4} \left(\frac{1}{\rho_- c_-^2} - \frac{1}{\rho_+ c_+^2} \right), \quad c_v = \frac{1}{4} (\rho_- - \rho_+), \quad c_t = \frac{c_v}{\omega^2 \rho_- \rho_+}.$$

The projected vertical force has fixed radial measure $F_i = \int 2\pi r B_i (c_p |P_\Gamma|^2 + c_v |u_n|^2 + c_t |g_t|^2) dr$, because $n_z dS = 2\pi r dr$. Therefore

$$\delta F_i = 2 \int 2\pi r B_i \operatorname{Re} \left(c_p \overline{P_\Gamma} \delta P_\Gamma + c_v \overline{u_n} \delta u_n + c_t \overline{g_t} \delta g_t \right) dr.$$

This supplies a route to the full discrete acoustic shape Jacobian using linear solves and differentiated forms. It is a derived implementation target, not a claim that an analytic tangent has been implemented or validated. Independent finite shape perturbations, numerical refinement and boundary limits are still required. The derivation does not add viscosity, streaming, thermal coupling or three-dimensional disturbances to the axisymmetric model.

A.6 Normalization and sign audit

The following dimensions provide checks for subsequent implementations.

Quantity	SI units
Ψ, W, R_o, R_i, η	m
$\kappa, \mathbf{H}_\Psi$	m^{-1}
$\sigma\kappa, \Delta\rho\Phi, \lambda_V, \Pi, \mathbf{R}$	Pa
$\mathbf{w}; \mathbf{W}$	$\text{m s}^{-1}; \text{m}^2 \text{s}^{-2}$
$\mathbf{K}; \mathbf{K}_i$	$\text{kg m}^{-3}; \text{kg m}^{-1}$
$F_i, b_i; \mathbf{R}_P$	N; kg s^{-1}
$E, K_f; P_{\text{in}}$	J; W
$J_W, J_d, J_s, J_{\text{spot}}, J; \ell$	dimensionless; s^{-1}

As a radiation-flux check, a locally progressive plane wave with propagation unit vector $\mathbf{d}$ obeys $\mathbf{V} = P\mathbf{d}/(\rho c)$. Equation (4.3) then gives $\mathbf{R} = |P|^2 \mathbf{d} \otimes \mathbf{d}/(2\rho c^2)$: the momentum flux has the sign of propagation. For the lens normal convention, the hydrostatic balance is $p^- - p^+ + \Pi = \sigma\kappa$, consistent with equation (2.10). Neither check uses a numerical solution or apparatus-specific values.

B Optional optical-performance diagnostics

These diagnostics concern a later optical-use problem. They do not replace the two-face geometric maximum in section 17; refractive-index modulation is not part of the present shape-only objective.

B.1 Optical objectives and a slope-sensitive identity

Let $\mathcal{P}$ be a two-dimensional entrance pupil, $\xi \in \mathcal{P}$ its ray label, and $\mathbf{X}_\Gamma(\xi)$ the ray's first intended intersection with the interface. Every admitted ray must reach the intended transmitting patch and have an unobstructed optical continuation. Let $d\mu(\xi)$ be a nonnegative normalized illumination measure, $\int_{\mathcal{P}} d\mu = 1$. Define $\bar{\Psi} = \int_{\mathcal{P}} \Psi(\mathbf{X}_\Gamma) d\mu$ and $W = \Psi(\mathbf{X}_\Gamma) - \bar{\Psi}$. With positive optical-path and angular tolerances W_{ref} and θ_{ref} , respectively,

$$J_W = \int_{\mathcal{P}} \frac{W^2}{W_{\text{ref}}^2} d\mu, \quad J_d = \int_{\mathcal{P}} \frac{\|\mathbf{P}_\Gamma(\mathbf{d}_t - \mathbf{d}_i)\|^2}{\theta_{\text{ref}}^2} d\mu. \quad (\text{B.1})$$

Removing the optical piston does not move either conjugate. Tangential Snell refraction gives the exact identity

$$n_i \mathbf{P}_\Gamma(\mathbf{d}_t - \mathbf{d}_i) = \nabla_\Gamma \Psi. \quad (\text{B.2})$$

A small-amplitude, rapidly varying path error can therefore give a large ray error. Near grazing transmission, tangential control alone is poorly conditioned for the full direction.

For a real image and a detector plane through $\mathbf{I}$ with unit normal $\mathbf{e}_i$, require $\mathbf{d}_t \cdot \mathbf{e}_i > 0$ and positive propagation distance. Then $\mathbf{Y} = \mathbf{X}_\Gamma + [(\mathbf{I} - \mathbf{X}_\Gamma) \cdot \mathbf{e}_i] \mathbf{d}_t / (\mathbf{d}_t \cdot \mathbf{e}_i)$. For a positive spot-radius tolerance,

$$J_{\text{spot}} = \int_{\mathcal{P}} \frac{|\mathbf{Y} - \mathbf{I}|^2}{r_{\text{ref}}^2} d\mu. \quad (\text{B.3})$$

Geometric ray errors are distinct from diffraction and measured optical quality.

B.2 Fast optical modulation

Let $\xi_a = \text{Re}(\Xi e^{-i\omega t})$ be normal acoustic displacement, and $\delta n_o, \delta n_i$ index perturbations along the ray segments $\mathcal{L}_o, \mathcal{L}_i$. For real conjugates and a nominal Fermat ray,

$$\delta\Psi = (n_o \mathbf{d}_o - n_i \mathbf{d}_i) \cdot \mathbf{n} \xi_a + \int_{\mathcal{L}_o} \delta n_o \, ds + \int_{\mathcal{L}_i} \delta n_i \, ds. \quad (\text{B.4})$$

Here s is arclength. Virtual conjugates use signed path segments. This is a first-order estimate, not a complete acousto-optic scattering calculation. Exposure time, index derivatives and the fast response determine its relevance. Physical interface vibration still belongs to the geometric error even when optical modulation is outside the experiment's objective.

References

- [1] N. Bertin, H. Chraïbi, R. Wunenburger, J.-P. Delville, and E. Brasselet. “Universal morphologies of fluid interfaces deformed by the radiation pressure of acoustic or electromagnetic waves”. In: *Physical Review Letters* 109 (2012), p. 244304. DOI: [10.1103/PhysRevLett.109.244304](https://doi.org/10.1103/PhysRevLett.109.244304).
- [2] H. Chesneau, H. Chraïbi, N. Bertin, J. Petit, J.-P. Delville, E. Brasselet, and R. Wunenburger. “Numerical simulation of universal morphogenesis of fluid interface deformations driven by radiation pressure”. In: *Physical Review E* 106 (2022), p. 065104. DOI: [10.1103/PhysRevE.106.065104](https://doi.org/10.1103/PhysRevE.106.065104).
- [3] F. Sisombat, T. Devaux, L. Haumesser, and S. Callé. “Contactless deformation of fluid interfaces by acoustic radiation pressure”. In: *Scientific Reports* 13 (2023), p. 14703. DOI: [10.1038/s41598-023-39464-0](https://doi.org/10.1038/s41598-023-39464-0).
- [4] J. H. Joergensen and H. Bruus. “Theory of pressure acoustics with thermoviscous boundary layers and streaming in elastic cavities”. In: *The Journal of the Acoustical Society of America* 149.5 (2021), pp. 3599–3610. DOI: [10.1121/10.0005005](https://doi.org/10.1121/10.0005005).
- [5] C. Huh and L. E. Scriven. “Hydrodynamic model of steady movement of a solid/liquid/fluid contact line”. In: *Journal of Colloid and Interface Science* 35.1 (1971), pp. 85–101. DOI: [10.1016/0021-9797\(71\)90188-3](https://doi.org/10.1016/0021-9797(71)90188-3).
- [6] T. Qian, X.-P. Wang, and P. Sheng. “A variational approach to moving contact line hydrodynamics”. In: *Journal of Fluid Mechanics* 564 (2006), pp. 333–360. DOI: [10.1017/S0022112006001935](https://doi.org/10.1017/S0022112006001935).
- [7] R. Finn. *Equilibrium Capillary Surfaces*. Vol. 284. Grundlehren der mathematischen Wissenschaften. New York: Springer, 1986. DOI: [10.1007/978-1-4613-8584-4](https://doi.org/10.1007/978-1-4613-8584-4).
- [8] M. Elgarisi, V. Frumkin, O. Luria, and M. Bercovici. “Fabrication of freeform optical components by fluidic shaping”. In: *Optica* 8.11 (2021), pp. 1501–1506. DOI: [10.1364/OPTICA.438763](https://doi.org/10.1364/OPTICA.438763).
- [9] J. S. Bach and H. Bruus. “Theory of pressure acoustics with viscous boundary layers and streaming in curved elastic cavities”. In: *The Journal of the Acoustical Society of America* 144.2 (2018), pp. 766–784. DOI: [10.1121/1.5049579](https://doi.org/10.1121/1.5049579).
- [10] K. Pieper, B. Q. Tang, P. Trautmann, and D. Walter. “Inverse point source location with the Helmholtz equation on a bounded domain”. In: *Computational Optimization and Applications* 77.1 (2020), pp. 213–249. DOI: [10.1007/s10589-020-00205-y](https://doi.org/10.1007/s10589-020-00205-y).
- [11] M. Hubenthal and D. Onofrei. “Sensitivity analysis for active control of the Helmholtz equation”. In: *Applied Numerical Mathematics* 106 (2016), pp. 1–23. DOI: [10.1016/j.apnum.2016.03.003](https://doi.org/10.1016/j.apnum.2016.03.003).
- [12] S. von Kowalevsky. “Zur Theorie der partiellen Differentialgleichungen”. In: *Journal für die reine und angewandte Mathematik* 80 (1875), pp. 1–32. DOI: [10.1515/crll.1875.80.1](https://doi.org/10.1515/crll.1875.80.1).
- [13] Z.-Q. Luo, W.-K. Ma, A. M.-C. So, Y. Ye, and S. Zhang. “Semidefinite relaxation of quadratic optimization problems”. In: *IEEE Signal Processing Magazine* 27.3 (2010), pp. 20–34. DOI: [10.1109/MSP.2010.936019](https://doi.org/10.1109/MSP.2010.936019).
- [14] N. J. A. Egarguin, D. Onofrei, and E. Platt. *Sensitivity analysis for the active manipulation of Helmholtz fields in 3D*. 2018. DOI: [10.48550/arXiv.1810.02407](https://doi.org/10.48550/arXiv.1810.02407). arXiv: [1810.02407](https://arxiv.org/abs/1810.02407).
- [15] M. B. Giles and N. A. Pierce. “An introduction to the adjoint approach to design”. In: *Flow, Turbulence and Combustion* 65 (2000), pp. 393–415. DOI: [10.1023/A:1011430410075](https://doi.org/10.1023/A:1011430410075).
- [16] F. Denner. “Frequency dispersion of small-amplitude capillary waves in viscous fluids”. In: *Physical Review E* 94 (2016), p. 023110. DOI: [10.1103/PhysRevE.94.023110](https://doi.org/10.1103/PhysRevE.94.023110).
- [17] S. Boyd, L. El Ghaoui, E. Feron, and V. Balakrishnan. *Linear Matrix Inequalities in System and Control Theory*. Vol. 15. Studies in Applied Mathematics. Philadelphia: Society for Industrial and Applied Mathematics, 1994. DOI: [10.1137/1.9781611970777](https://doi.org/10.1137/1.9781611970777).
- [18] A. Silva-Lora and R. Torres. “Superconical aplanatic ovoid singlet lenses”. In: *Journal of the Optical Society of America A* 37.7 (2020), pp. 1155–1165. DOI: [10.1364/JOSAA.392795](https://doi.org/10.1364/JOSAA.392795).